Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Signal and Information Processing Laboratory
Institut für Signal- und Informationsverarbeitung

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Bounds on the Capacity of Free-Space Optical Intensity Channels

Michèle Wigger

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Advisor: Stefan M. Moser

Co-Advisor: Daniel Hösli

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Abstract

Upper and lower bounds on the capacity of the free-space optical intensity channel are computed. The lower bounds are found using the common expression for the capacity and the power entropy inequality, whereas the upper bounds are mainly found with a recently presented technique from Lapidoth and Moser using the dual expression for the capacity. At high power upper and lower bounds are presented which become asymptotically tight. Also for low power some good upper and lower bounds for the capacity are found. Most of the results are new and cannot be compared to previously found expressions for the capacity. However, where this is possible more general results are obtained in this report.

Keywords: channel capacity, upper and lower bounds, free-space optical intensity channel, infrared channel

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Chapter 1

Introduction

Optical links through air are often used in wireless communication systems in- and outside of buildings. The advantages of such links are mainly its high data rate and its low implementation costs in comparison to wired communication (e.g. through optical fibers). Additionally, they are robust against electromagnetic disturbances and interference with most other wireless systems. Typically the wavelength of the transmitted light lies in the near infrared range (ca. 700nm - 1000nm). These communication links differ from common radio communication especially in the fact that the input of the channel is proportional to the intensity of the transmitted light and not to the light's amplitude. Therefore, the input of the channel is restricted to non-negative values and any power constraint (for this channel we have average and peak power constraints) affects directly the input and not the square of the input.

The considered channel can be modelled by an Additive White Gaussian Noise channel with an input that is constrained to be non-negative and to have a limited peak and average value. For reasons of simplicity in this report two simplified models are considered as well, namely by relaxing either the average or the peak constraint.

Channel capacity is one of the most important properties of a communication channel. It represents the maximal rate at which data can be transmitted with arbitrary small probability of error. For most channels the capacity is not known, as it is very difficult to compute analytically exact. The free-space optical intensity channel is such a channel where only some upper bounds or asymptotic capacity results for a simplified channel model

are known. The objective of this report is to find good lower and upper bounds on the capacity at low and at high power.

The derivation of the lower bounds is based on the common formula for the capacity, which requires a maximization over the space of input densities. Choosing an appropriate input density, which not necessarily achieves the maximum, combined with the entropy power inequality leads to good lower bounds. For the upper bounds a technique proposed in [3] is used. There a dual expression for the capacity is considered which contains a minimization over the space of output densities. Therefore, an appropriate choice of density at the output very naturally yields an upper bound. The bounds presented in this thesis become tight for high power. The results for the simplified models with relaxed either average or peak power constraint will be shown to be special cases for the complete model with peak and average power constraints.

The information theoretical formulae needed to understand this report are listed in Chapter 2. Chapter 3 explains the physical background of the free-space optical intensity channel. From this explanation a mathematical model is derived, which can be used to compute upper and lower bounds. These bound are presented in the following two chapters according to whether they lead to asymptotically good results for high power (Chapter 4) or for low power (Chapter 5). The bounds are then discussed all together in Chapter 6 in order to get an overview over the behaviour of the capacity in all power regions. Additionally, some older results are compared to the received asymptotic expressions for the capacity at high power. In Chapter 7 the results are summarized and some outlook is given on possible future research topics.

Chapter 2

Channel Capacity and Duality

In this chapter the concepts of a channel and the channel capacity are introduced. Furthermore, the information theoretical background is given to compute good upper and lower bounds on the channel capacity.

First, the common expression for the capacity is presented. Then, the dual expression is introduced and a method to compute upper bounds is given. Next, the entropy power inequality is presented and last it is shown which densities maximize differential entropy under some constraints. These two later concepts will especially be used in the computation of lower bounds for the capacity.

2.1 Channel Capacity

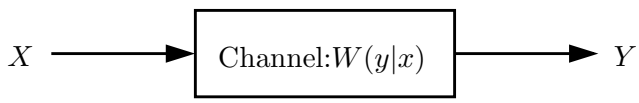

Figure 2.1: Block diagram of a communication channel

A physical channel is a communication device with an input and an output. The channel output is in general not fully determined by the input, but depends as well on some noise. So if a signal is introduced at the input of the channel some other signal is produced at the channel output. To use such a device for communication the receiver has to decide from the output

about which signal has been sent. If the decision does not match with the input, an error in the communication has occurred.

A mathematical analog to the physical channel is shown in Figure 2.1. Following the definition in [4] a continuous memoryless channel is a device where an input X induces an output Y, where Y is a random variable with probability density obeying the channel law $W(y|x)$. The channel law considers the noise introduced to the channel and characterizes the channel. For the channel we are going to analyze in this report the output depends only on the actual input, therefore the channel is called memoryless. As the output is a random variable, two different inputs can produce the same output! At the output a decision following a certain rule has to be taken. If this decision is different from the input, then an error in the communication has occurred.

The input x of the channel is a value in the alphabet $\mathcal{X}$, whereas the channel output y lies in the alphabet $\mathcal{Y}$. In n consecutive channel uses an entire codeword $\mathbf{x} = (x_1, ..., x_n)$ is transmitted, where the alphabet of $\mathbf{x}$ is usually restricted to a chosen codebook $\mathcal{C}$. Every codebook for n channel uses fulfills the condition:

$$\mathcal{C} \subseteq \mathcal{X}^n. \tag{2.1}$$

The alphabet of possible outputs $\mathbf{y} = (y_1, ..., y_n)$ is in general $\mathcal{Y}^n$ and the decision taken at the output about which codeword has been sent follows a chosen mapping

$$\Phi_n(y) : \mathcal{Y}^n \to \mathcal{C}. \tag{2.2}$$

The chosen codebook $\mathcal{C}$ determines the rate, which is defined as

$$R = \frac{\log |\mathcal{C}|}{n} \tag{2.3}$$

and is a measure of how many different symbols can be sent per channel use. A rate is said to be achievable for a certain channel if a codebook $\mathcal{C}$ and a mapping $\Phi_n(y)$ can be chosen in such a way that the probability of error in the communication for every possibly sent codeword $\mathbf{x} \in \mathcal{C}$ tends to 0 for when n tends to infinity. The probability of error given the codeword $\mathbf{x}$ has been sent is:

$$\mathrm{P}(\text{error}|\mathbf{x}) = \sum_{\substack{y \in \mathcal{Y}^n, \\ \Phi_n(y) \neq \mathbf{x}}} \prod_{i=1}^{n} W(y_i|x_i). \tag{2.4}$$

The maximum of achievable rates of a channel is called the channel capacity. It can be shown [4], that this definition is equal to defining the capacity C as

$$C = \max_{Q \in \mathcal{P}(\mathcal{X})} I(Q, W), \tag{2.5}$$

where $\mathcal{P}(\mathcal{X})$ is the set of all probability densities over $\mathcal{X}$ and $I(Q, W)$ denotes the mutual information between the input, when the transmitted symbols have been chosen according to the probability density $Q(x)$, and the corresponding output induced by the channel with transition probability $W(y|x)$. The mutual information between the input and the output of a continuous channel is defined as:

$$I(Q, W) = \int_{x \in \mathcal{X}} \int_{y \in \mathcal{Y}} Q(x) W(y|x) \log \frac{W(y|x)}{(QW)(y)} \, \mathrm{d}y \, \mathrm{d}x, \tag{2.6}$$

where $(QW)(\cdot)$ denotes the output probability density $R(y)$ induced by the channel with law $W(y|x)$ and an input chosen according to the probability density $Q(x)$:

$$R(y) = (QW)(y) = \int_{x' \in \mathcal{X}} Q(x') W(y|x') \, \mathrm{d}x', \qquad y \in \mathcal{Y}. \tag{2.7}$$

The mutual information of continuous random variables X and Y can as well be expressed in terms of the differential entropy of the output

$$h(Y) = -\int_{y \in \mathcal{Y}} R(y) \log R(y) \, \mathrm{d}y, \tag{2.8}$$

where $R(y)$ is defined as in 2.7, and the conditional differential entropy of the channel

$$\begin{aligned} h(Y|X) &= \int_{x \in \mathcal{X}} Q(x) h(Y|X = x) \, \mathrm{d}x \\ &= -\int_{x \in \mathcal{X}} Q(x) \int_{y \in \mathcal{Y}} W(y|x) \log W(y|x) \, \mathrm{d}y \, \mathrm{d}x. \end{aligned} \tag{2.9}$$

Then the mutual information between the input and the output of a channel can be written as

$$I(Q, W) = h(Y) - h(Y|X) \tag{2.10}$$

and with (2.5) we find a new expression for the capacity

$$C = \max_{Q \in \mathcal{P}(\mathcal{X})} \Big\{ h(Y) - h(Y|X) \Big\}. \tag{2.11}$$

2.2 Dual Expression of the Channel Capacity

Additionally to the formulae of Section 2.1 there exists a dual expression for the capacity. In order to introduce this dual expression we define the relative entropy of two probability densities $f(\cdot)$ and $g(\cdot)$

$$D(f(\cdot)||g(\cdot)) = \int_{-\infty}^{\infty} f(x) \log \frac{f(x)}{g(x)} \, \mathrm{d}x, \tag{2.12}$$

where the support set of $f(\cdot)$ has to be contained in the support set of $g(\cdot)$. Then, the dual expression for the capacity is [5]:

$$C = \min_{R \in \mathcal{P}(\mathcal{Y})} \max_{x \in \mathcal{X}} D(W(\cdot|x)||R(\cdot)). \tag{2.13}$$

Unlike in Section 2.1 $R(y)$ does not necessarily have to be induced by the channel from some input density, but can be any probability density on the output alphabet $\mathcal{Y}$.

The advantage of (2.13) is that it contains a minimization over all the output densities instead of a maximization over all the input densities as in (2.5). Therefore, for every choice of probability density $R(y)$ an upper bound for the capacity is obtained:

$$C \leq \max_{x \in \mathcal{X}} D(W(\cdot|x)||R(\cdot)), \tag{2.14}$$

where we still have to maximize over all $x \in \mathcal{X}$. But maximizing over all the elements in $\mathcal{X}$ is much easier than maximizing over the set of all probability densities.

Nevertheless, expression (2.14) is not used in this report, but a similar expression for the capacity is preferred. We want to introduce this new expression next. We can write the mutual information of a channel as [6]:

$$I(Q, W) = \int_{x \in \mathcal{X}} Q(x) D(W(\cdot|x)||R(\cdot)) \, \mathrm{d}x - D((QW)(\cdot)||R(\cdot)). \tag{2.15}$$

Considering (2.5) and (2.15) and the fact that the relative entropy is always nonnegative [4], a new upper bound for the capacity is obtained:

$$C = \max_{Q \in \mathcal{P}(\mathcal{X})} I(Q, W) \leq \max_{Q \in \mathcal{P}(\mathcal{X})} \int_{x \in \mathcal{X}} Q(x) D(W(\cdot|x)||R(\cdot)) \, \mathrm{d}x. \tag{2.16}$$

If we denote the maximizing input density by Q^* we get the final upper bound which we are going to use later on:

$$\begin{aligned} C &\leq \mathrm{E}_{Q^*} \left[D(W(\cdot|X)||R(\cdot)) \right] \\ &= \mathrm{E}_{Q^*} \left[- \int_{y \in \mathcal{Y}} W(y|X) \log R(y) \, \mathrm{d}y \right] - h_{Q^*}(Y|X). \end{aligned} \tag{2.17}$$

From (2.15) and (2.16) we can see that equality in (2.17) is reached if and only if $R(y)$ corresponds to the capacity achieving output density $R^*(y)$ induced by the input density $Q^*(x)$.
The advantage of (2.17) is that it leads to an upper bound for the capacity for any arbitrary output density $R(y)$. Therefore we can choose a $R(y)$ that leads to a mathematically tractable upper bound.

2.3 Maximum Differential Entropy Distributions

In Chapters 4 and 5 we will need to know the probability density over a support set $\mathcal{S}$, which respects some moment constraints and especially maximizes the differential entropy.

Theorem 2.1 *Let X be a random variable in the support set $\mathcal{S}$ with density $f(x)$ fulfilling the constraints $\mathrm{E}[r_j(x)] = A_j$, $j = 1, \ldots, m$. Then the differential entropy $h(X)$ is maximized if and only if $f(x)$ is of the form:*

$$f(x) = e^{\lambda_0 + \sum_{j=1}^{m} \lambda_j r_j(x)}, \tag{2.18}$$

where $\lambda_0, \ldots, \lambda_m$ have to be chosen such that $f(x)$ is a probability density over $\mathcal{S}$, meaning that it integrates to 1, and the moment constraints are fulfilled.

Proof: See [4] □

With the help of this theorem we want to derive the density function of a random variable defined in some support set $\mathcal{S} = [a, b]$ and respecting an average constraint, which maximizes the differential entropy.

Theorem 2.2 *Let X be a random variable in the support set $\mathcal{S} = [a, b]$ with average constraint $\mathrm{E}[X] = P_{\mathrm{avg}}$ and density $f(x)$. Then the differential entropy is maximized if and only if*

$$f(x) = \frac{\lambda}{e^{\lambda b} - e^{\lambda a}} e^{\lambda x}, \tag{2.19}$$

where λ is defined by:

$$-\frac{1}{\lambda} + \frac{e^{\lambda b} b - e^{\lambda a} a}{e^{\lambda b} - e^{\lambda a}} = P_{\mathrm{avg}}. \tag{2.20}$$

Proof: Using Theorem 2.1 we get the following expression:

$$f(x) = e^{\lambda_0 + \lambda_1 x} = K e^{\lambda x}, \tag{2.21}$$

where $f(x)$ has to fulfill the conditions:

$$\int_a^b f(x)\mathrm{d}x = \int_a^b K e^{\lambda x}\mathrm{d}x = \frac{K}{\lambda}\left(e^{\lambda b} - e^{\lambda a}\right) = 1 \tag{2.22}$$

$$\begin{aligned}\int_a^b f(x)x\mathrm{d}x &= \int_a^b K e^{\lambda x} x\mathrm{d}x \\ &= \frac{K}{\lambda}\left(e^{\lambda b}b - e^{\lambda a}a\right) - \frac{K}{\lambda^2}\left(e^{\lambda b} - e^{\lambda a}\right) = P_{\text{avg}}\end{aligned} \tag{2.23}$$

Examining (2.22) gives us an expression for K:

$$K = \frac{\lambda}{e^{\lambda b} - e^{\lambda a}}. \tag{2.24}$$

Introducing this expression in (2.23) yields:

$$-\frac{1}{\lambda} + \frac{e^{\lambda b}b - e^{\lambda a}a}{e^{\lambda b} - e^{\lambda a}} = P_{\text{avg}}. \tag{2.25}$$

So the probability density over $\mathcal{S} = [a, b]$ respecting an average constraint, which maximizes the differential entropy, is the density corresponding to an exponential distribution, where the parameter λ depends on the average and the support set. □

Corollary 2.3 *If for the probability density found in Theorem 2.2 the average constraint $P_{\text{avg}} \leq \frac{b-a}{2}$, then the parameter λ is non-positive.*

Proof: If λ is positive, then $f(x)$ is strictly monotonically increasing and hence $\mathrm{E}[X] = P_{\text{avg}}$ is larger than $\frac{b-a}{2}$. So we have found that λ has to be nonpositive in order to obtain an average less than or equal to $\frac{b-a}{2}$. □

Next, we search for the probability density over the support set $\mathcal{S} = [a, \infty]$ considering a finite average constraint which maximizes differential entropy. As P_{avg} stays finite, Corollary 2.3 implies that λ is nonpositive. Using Theorem 2.2 we find the maximizing density:

$$f(x) = \lim_{b\to\infty} \frac{\lambda}{e^{\lambda b} - e^{\lambda a}} e^{\lambda x} = \lim_{b\to\infty} \frac{\tilde{\lambda}}{e^{-\tilde{\lambda} a} - e^{-\tilde{\lambda} b}} e^{-\tilde{\lambda} x} = \frac{\tilde{\lambda}}{e^{-\tilde{\lambda} a}} e^{-\tilde{\lambda} x} \tag{2.26}$$

where we introduced $\tilde{\lambda} = -\lambda$ and the last equality follows from the fact that $\tilde{\lambda} \geq 0$.
The Condition (2.25) becomes:

$$\begin{aligned} P_{\text{avg}} &= \lim_{b\to\infty}\left\{-\frac{1}{\lambda}+\frac{e^{\lambda b}b-e^{\lambda a}a}{e^{\lambda b}-e^{\lambda a}}\right\} = \lim_{b\to\infty}\left\{\frac{1}{\tilde{\lambda}}+\frac{e^{-\tilde{\lambda} b}b-e^{-\tilde{\lambda} a}a}{e^{-\tilde{\lambda} b}-e^{-\tilde{\lambda} a}}\right\} \\ &= \frac{1}{\tilde{\lambda}}+a. \end{aligned} \tag{2.27}$$

We can state these results in a corollary:

Corollary 2.4 *Given a density function $f(x)$ over the support set $\mathcal{S} = [a,\infty]$ respecting the average constraint $\mathrm{E}[X] = P_{\text{avg}}$. Then $f(x)$ maximizes the differential entropy if and only if*

$$f(x) = \frac{1}{P_{\text{avg}}-a}e^{-\frac{x-a}{P_{\text{avg}}-a}}. \tag{2.28}$$

Proof: Insert (2.27) into (2.26). □

Last, we want to find the probability density over the support set $\mathcal{S} = [a,b]$ without any further constraints which maximizes differential entropy. This means that in (2.19) we have to set $\lambda = 0$:

$$f(x) = \lim_{\lambda\to 0}\frac{\lambda}{e^{\lambda b}-e^{\lambda a}}e^{\lambda x} = \lim_{\lambda\to 0}\frac{\lambda}{e^{\lambda b}-e^{\lambda a}} = \lim_{\lambda\to 0}\frac{1}{e^{\lambda b}b-e^{\lambda a}a} = \frac{1}{b-a}, \tag{2.29}$$

where the third equality follows from Bernoulli-De l'Hôpital's rule.

Corollary 2.5 *The probability density $f(x)$ over a support set $\mathcal{S} = [a,b]$, where no further moment constraints have to be fulfilled, maximizes the differential entropy if and only if it is a uniformly distributed density function:*

$$f(x) = \frac{1}{b-a}. \tag{2.30}$$

2.4 Input and Output Entropy

Very often it is difficult to find even some lower bounds for the channel capacity, as the differential entropy $h(Y)$ has to be computed for an output density $R(y)$ which corresponds to some possible input density. It is easier to compute the differential entropy of some input density $h(X)$ because all the input densities are valid. Therefore, a relationship between the input

entropy and the output entropy of the cannel would be very helpful to us. And indeed we can make use of the following theorem:

Theorem 2.6 (Entropy Power Inequality) *If X and Z are independent random variables with densities, then*

$$e^{2h(X+Z)} \geq e^{2h(X)} + e^{2h(Z)}. \tag{2.31}$$

Proof: See [4]. □

In the case of an additive noise channel X may correspond to the input and Z to the added noise. Then $X + Z$ corresponds to the channel output Y, i.e.

$$h(X + Z) = h(Y). \tag{2.32}$$

In Chapter 4 it suffices to consider a weaker lower bound. We derive from Theorem 2.31:

Corollary 2.7 *If X and Z are independent random variables with densities, then*

$$h(X + Z) \geq h(X). \tag{2.33}$$

Proof: Insert the inequality $e^{h(Z)} \geq 0$ in (2.31). □

Chapter 3

Channel Model

In this chapter a mathematical model for the free-space optical intensity channel is presented. This model will be used in Chapters 4 and 5 to compute lower and upper bounds for the capacity.

First, the channel is described physically and then in the second section a mathematical model is developed from this description. For the further use this model is too complicated and therefore a simplification is presented.

With this simplified model we are able to compute the conditional differential entropy and some other statistical measures of the free-space optical intensity channel as shown in the last section of this chapter.

3.1 Physical Description

In the free-space optical intensity channel (see Figure 3.1) the transmission of the input signal is performed by some light emitting diodes (LED) or laser diodes (LD). Conventional diodes emit light of wavelength between 850 and 950 nm, this means in the infrared area. The modulation of the signals onto this infrared light is technically very difficult using amplitude or frequency modulation [7], as used for example in radio communication. Instead the simplest way is to modulate the signal onto the optical intensity of the emitted light, which is proportional to the optical power. Using intensity modulation (IM) the intensity of the emitted light is proportional to the electrical input current $X(t)$:

$$I_{\text{out}}(t) = gX(t), \tag{3.1}$$

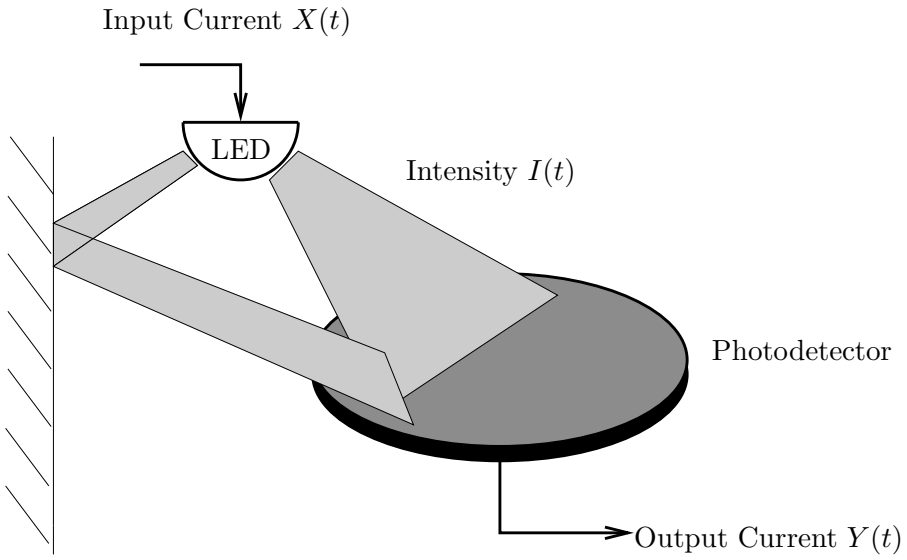

Figure 3.1: Physical channel model

where g is the optical gain of the diode in $\frac{W}{Am^2}$. Therefore, the instantaneous optical power is proportional to the electrical input current $X(t)$ and not to its square as this is usually the case for radio communication.

At the receiver direct detection (DD) of the incident optical intensity is performed. This means that the photodetector produces an electrical current $Y(t)$ at the output which is proportional to the detected intensity:

$$Y(t) = rI_{\text{in}}(t), \tag{3.2}$$

where r is the detector sensitivity in $\frac{Am^2}{W}$.

In general the emitted intensity from the LED I_{out} is not equal to the incident intensity at the photodetector I_{in}, but the different paths of propagation of the light have to be considered. Additionally, the transmission of the signal is corrupted by some noise. The dominant noise source for the optical intensity is strong ambient light. Even if optical filters are used to filter the ambient light, some part of it always arrives at the photodetector, and has typically much larger power than the actual signal. This effect causes high intensity shot noise in the received electrical signal $Y(t)$. A further important noise source that affects the electrical output signal is thermal noise in the pre-amplifier of the photodetector.

From the properties of the free-space optical intensity channel some restrictions to the electrical input signal $X(t)$ arise. As the optical intensity

$I_{\text{out}}(t)$ cannot be negative and is proportional to the electrical input signal, it follows that

$$X(t) \geq 0. \tag{3.3}$$

Due to eye safety and thermal skin damage the optical peak power and the optical average power have to be constrained. Considering (3.1), peak and average constraints arise as well for the electrical input signal:

$$\begin{aligned} \mathrm{E}[X] &\leq P_{\text{avg},E} \quad \text{and} && (3.4) \\ X(t) &\leq A_E. && (3.5) \end{aligned}$$

We will see in the following section that in contrast to $X(t)$ the electrical output signal $Y(t)$ can very well be negative, as the noise introduced at the receiver is not necessarily positive.

3.2 Mathematical Model

From the physical description of Section 3.1 we have to find a mathematical model that we can use in the following chapters to compute the bounds on the channel capacity.

First, we want to find a relationship between I_{out} and I_{in}, the intensity of the light emitted from the LED and the intensity measured by the photodetector. If we do not consider the intensity added by the ambient light, which we will take into account together with the other sources of noise at the electrical signal $Y(t)$, then there is a linear relation between the two intensities [7]. This relation can be characterized by the impulse response $h(t)$ of the channel:

$$I_{\text{in}}(t) = I_{\text{out}}(t) * h(t), \tag{3.6}$$

where $h(t)$ considers the different paths of propagation of the infrared light from the LED to the photodetector.

Next, we have to take into account the dominant noise sources. As already mentioned we will take care of these combined at the received signal $Y(t)$. There we can make use of the central limit theorem to approximate the principal sources of noise, the high intensity shot noise and the thermal noise, by Additive White Gaussian Noise (AWGN) as all sources are independent

of each other and of the signal. We now get the following mathematical model for the channel, shown as well in Figure 3.2:

$$Y(t) = grX(t) * h(t) + Z(t), \tag{3.7}$$

where the noise $Z(t)$ is zero mean Gaussian with variance σ^2, i.e. its density is given by

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}}. \tag{3.8}$$

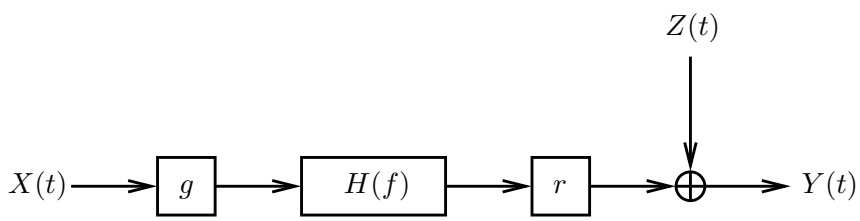

Figure 3.2: Mathematical channel model

3.2.1 Simplified Model

The channel model found so far is too complicated to compute lower and upper bounds for the channel capacity. Therefore we will make some further simplifications.

We assume that the intensity of the light propagating along the direct line of sight path is dominant. This is a good approximation if there is a direct line of sight between the emitting LED and the receiving photodetector, see [7] and [1]. Then the impulse response of the channel is approximately:

$$h(t) = \delta(t). \tag{3.9}$$

Further we can assume without loss of generality that the responsivity r and the gain g are equal to 1. Hence we get a simplified channel model (see Figure 3.3) which is characterized as follows:

$$Y(t) = X(t) + Z(t), \tag{3.10}$$

where again $Z(t)$ is zero mean Gaussian with variance σ^2. In this simplified model the constraints on the electrical input signal (3.3), (3.4) and (3.5)

become:

$$0 \leq X(t) \leq A, \tag{3.11}$$

$$\mathrm{E}[X] \leq P_{\mathrm{avg}}, \tag{3.12}$$

where A is the peak optical intensity constraint and P_{avg} denotes the average optical intensity constraint. Remember that the intensity is proportional to the optical power. Therefore we will talk henceforth about power constraints rather than about intensity constraints, which seems to be more intuitive.

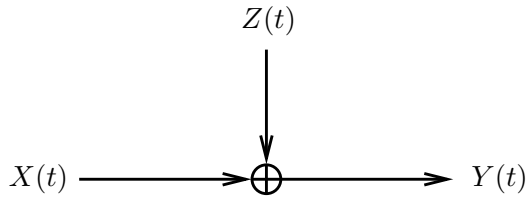

Figure 3.3: Simplified mathematical channel model

This simplified model is not only used for the free-space optical intensity channel, but as well for fiber-optical channels with negligible dispersion and signal independent Additive White Gaussian Noise [8].

3.3 Statistical Measures of the Channel

We can use now the obtained channel model (3.10) to compute some important statistical measures for the free-space optical intensity channel. First, we investigate the transition probability of the channel, meaning the conditional probability density of the output given some input:

$$W(y|x) = f_Z(y-x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}}. \tag{3.13}$$

We can see that the channel is memoryless and stationary. Therefore, in the remaining we will not consider the time dimension any longer.

In the following chapters we will have to know the relation between the expectation of the input of the channel and the expectation of the output.

Lemma 3.1 *For the channel model (3.10) with 0 mean noise $Z(t)$ holds:*

$$\mathrm{E}[Y] = \mathrm{E}[X]. \tag{3.14}$$

Proof: Considering that the noise $Z(t)$ has 0 mean gives

$$\mathrm{E}[Y] = \mathrm{E}[X+Z] = \mathrm{E}[X] + \mathrm{E}[Z] = \mathrm{E}[X]. \tag{3.15}$$

□

3.3.1 Conditional Differential Entropy of the Output

In view of (2.11) and (2.17) we need to know the conditional differential entropy of the output to compute both some upper and some lower bound for the channel capacity. With (3.13) we can find the conditional differential entropy of the output for this channel, which, as we will see, is independent of the input density.

Theorem 3.2 *For a channel with transition probability*

$$W(y|x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}} \tag{3.16}$$

the conditional differential entropy of the output is given by:

$$h(Y|X) = \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{3.17}$$

Proof:

$$h(Y|X) = \int_0^A Q(x)\ h(Y|X=x)\,\mathrm{d}x, \tag{3.18}$$

where

$$\begin{aligned} h(Y|X=x) &= -\int_{-\infty}^{\infty} W(y|x)\log W(y|x)\,\mathrm{d}y \\ &= -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}} \log\left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}}\right)\mathrm{d}y \\ &= \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{3.19}$$

□

3.3.2 Concavity of the Average Power Constrained Channel

The channel of interest is subject to an average power constraint (3.12). From [9] we know that the capacity of a discrete memoryless channel with a moment constraint is a non-decreasing concave function of this constraint. As long as the unconstrained capacity is not achieved, the capacity is even

strictly increasing. This means that for the capacity achieving input density the constraint is satisfied with equality. We assume that for sufficiently smooth input densities this holds as well for continuous channels. So when in the next chapters we talk about an average constraint for the input, it will imply that the average of the capacity achieving input density is exactly equal to this constraint.

3.3.3 Symmetry of the Only Peak Power Constrained Channel

In this chapter we analyze the free-space optical intensity channel where the average power constraint is relaxed. Then the input is subject to only a peak and a non-negativity constraint. For this simplified channel model we want to find the expected value of the input density which is capacity achieving. This expectation will be used in the computation of upper bounds for the capacity in Chapters 4 and 5.

Theorem 3.3 *The expected value of the capacity achieving input density $Q^*(x)$ of the free-space optical intensity channel subject to only a peak power constraint is equal to half the peak:*

$$\mathrm{E}_{Q^*}[X] = \frac{A}{2} \tag{3.20}$$

Proof: From [10] we know that the capacity achieving input density is unique. We will show that if an input density has expectation not equal to half the peak, then a different input density can be found, which yields the same mutual information between the input and the output of the channel. Therefore we can conclude that such an input density cannot be capacity achieving.

Assume that the capacity achieving input density $Q^*(x)$ has an expected value $\mathrm{E}_{Q^*}[X]$ which is larger than $\frac{A}{2}$. Then we can construct a different input distribution $\tilde{Q}(x)$ which is symmetric to $Q^*(x)$ relative to $\frac{A}{2}$ and therefore has expectation $\mathrm{E}_{\tilde{Q}}[X] = A - \mathrm{E}_{Q^*}[X]$. Both of these distributions have the same entropy:

$$h_{Q^*}(X) = h_{\tilde{Q}}(X). \tag{3.21}$$

Because of the symmetry of the AWGN channel $Q^*(x)$ and $\tilde{Q}(x)$ induce output densities which again are symmetric respective to $\frac{A}{2}$:

$$R_{\tilde{Q}}(y) = R_{Q^*}(A - y), \tag{3.22}$$

and therefore have the same differential entropy:

$$h_{Q^*}(Y) = h_{\tilde{Q}}(Y). \tag{3.23}$$

Thus, by (2.11) the input densities $Q^*(x)$ and $\tilde{Q}(x)$ yield the same mutual information of the channel. This contradicts the assumption that $Q(x)$ is the unique capacity achieving output density and therefore $\mathrm{E}_{Q^*}[X] > \frac{A}{2}$ is not possible. The same proof can be used to show that $\mathrm{E}_{Q^*}[X] < \frac{A}{2}$ is not possible either. Therefore, we can conclude that the capacity of the channel with only peak power constraint has to be reached by an input density with expectation value $\mathrm{E}_{Q^*}[X] = \frac{A}{2}$. □

Using the convexity of the capacity under a moment constraint stated in Section 3.3.2 it follows that an average power constraint makes only sense if the constraint is smaller than half the peak power. Otherwise the constraint is not fulfilled with equality and the channel behaves as there was no average power constraint.

Chapter 4

The Channel Capacity at High Power

In this chapter we are going to present the channel capacity of the free-space optical intensity channel at high optical power (average or peak). We are going to use the simplified channel model found in Section 3.2.1. For this model three different cases are distinguished. First we are going to compute bounds for the capacity of a channel respecting only the average power constraint. Next we will take into account only the peak power constraint and last we will present an upper and a lower bound for the capacity of the free-space optical intensity channel respecting both power constraints.

In all the cases the upper and lower bounds are asymptotically tight, meaning that their difference vanishes as the peak or average power tend to infinity. Furthermore, by relaxing one of the constraints in the asymptotic expression for the capacity the case of either only an average power or only a peak power constraint can directly be derived.

We will show that actually the expressions for the capacity at high power considering only one of the constraints are very useful also for practical reasons. We find closed-form expressions for the error that arises from not considering the non-dominant constraint, and we show that this error is quite small.

4.1 Introduction

In each of the following three sections lower bounds and upper bounds on the capacity are computed for one of the different channel models. In section 4.2 the peak power constraint has not been considered, whereas in Section 4.3 the average power constraint was not taken into account. Then, in Section 4.4, a result for the capacity considering both constraints is obtained.

The lower bounds are computed with (2.11) and the weaker version of the entropy power inequality (2.33), whereas the upper bounds are found with the help of the dual expression for the capacity (2.17).

4.2 Results with Average Power Constraint only

In this section the capacity of the free-space optical intensity channel with only average optical power constraint is discussed. We use the mathematical formulae from Chapter 2 to compute the lower and upper bounds, which, as we will see, become tight at high average power.

4.2.1 Lower Bound

We are going to compute a lower bound for the channel capacity with average power constraint $\mathrm{E}[X] \leq P_{\text{avg}}$. Using (2.33) to lower bound (2.11) we get:

$$C \geq h(X) - h(Y|X), \tag{4.1}$$

where the inequality is fulfilled for any arbitrary probability density at the input respecting the average power constraint. However, the best lower bound is achieved by maximizing $h(X)$, as $h(Y|X)$ is constant (see Theorem 3.2). From Corollary 2.4 we know that the probability density which maximizes differential entropy under an average and a non-negativity constraint is the density of the exponential distribution:

$$Q(x) = \frac{1}{P_{\text{avg}}} e^{-\frac{x}{P_{\text{avg}}}}, \quad x \in [0, \infty). \tag{4.2}$$

The corresponding entropy is:

$$\begin{aligned} h(X) &= -\int_0^\infty \frac{1}{P_{\text{avg}}} e^{-\frac{x}{P_{\text{avg}}}} \log\left(\frac{1}{P_{\text{avg}}} e^{-\frac{x}{P_{\text{avg}}}}\right) \mathrm{d}x \\ &= \log\left(P_{\text{avg}}\right) + 1. \end{aligned} \tag{4.3}$$

Thus, we have found a lower bound for the capacity with average power constraint P_{avg}:

$$C \geq C_{\text{L}} = \log\left(P_{\text{avg}}\right) + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{4.4}$$

4.2.2 Upper Bound

In the last section we found a lower bound for the channel capacity. Now we want to obtain an upper bound that is close to the lower bound at high average power. To compute the upper bounds we will use the dual expression for the capacity (2.17). Then two different probability densities will yield upper bounds which tend to the same asymptotic expression for high average power.

Fisher-Tippett Distribution

First, we compute an upper bound with the probability density of the Fisher-Tippett distribution at the output:

$$R(y) = \frac{1}{b}e^{\frac{y-a}{b} - \exp\frac{y-a}{b}}, \quad \forall y \in \mathbb{R}. \tag{4.5}$$

Applying this density to (2.17) we get the following upper bound for the capacity:

$$\begin{aligned} C \leq C_{\text{U,FT}} \quad = \quad & \frac{2(P_{\text{avg}} + \log P_{\text{avg}})}{P_{\text{avg}} + \log P_{\text{avg}} + \sqrt{(P_{\text{avg}} + \log P_{\text{avg}})^2 + 4\sigma^2}} \\ & + \log\left(\frac{\mathrm{P}_{\text{avg}} + \log P_{\text{avg}} + \sqrt{(P_{\text{avg}} + \log P_{\text{avg}})^2 + 4\sigma^2}}{2}\right) \\ & + \frac{2\sigma^2}{\left(P_{\text{avg}} + \log P_{\text{avg}} + \sqrt{(P_{\text{avg}} + \log P_{\text{avg}})^2 + 4\sigma^2}\right)^2} \\ & + \frac{1}{P_{\text{avg}}} - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{4.6}$$

For a derivation see Appendix A.1.

Double Sided Exponential Distribution

Next we will use the density of a double sided exponential distribution at the output:

$$R(y) = \begin{cases} \frac{p}{\beta}e^{-\frac{y-m}{\beta}} & y \geq m, \\ \frac{1-p}{\alpha}e^{\frac{y-m}{\alpha}} & y < m. \end{cases} \tag{4.7}$$

with parameters $m \leq 0$, $\alpha, \beta \geq 0$, $0 \leq p \leq 1$.

Applying again the dual expression to compute an upper bound for the capacity we obtain:

$$\begin{aligned} C \leq C_{\mathrm{U,Exp}} &= \log\left(\frac{\sigma}{\sqrt{2\pi}} e^{-\frac{(\log P_{\mathrm{avg}})^2}{2}} + P_{\mathrm{avg}} + \sigma \log P_{\mathrm{avg}}\right) \\ &\quad - \log\left(1 - \frac{1}{\log P_{\mathrm{avg}}}\right) + 1 + \frac{\sigma \log P_{\mathrm{avg}}}{\sqrt{2\pi}} e^{-\frac{(\log P_{\mathrm{avg}})^2}{2}} \\ &\quad - \frac{1}{2} \log\left(2\pi e \sigma^2\right), \end{aligned} \tag{4.8}$$

where

$$P_{\mathrm{avg}} \geq \max\left\{e, e^{\frac{1}{\sigma}}\right\}, \tag{4.9}$$

where the $\mathcal{Q}(\cdot)$-function is defined in the Appendix D.1.
For a derivation see Appendix A.2.

4.2.3 Capacity Results

In the previous two sections we have found some upper and lower bounds for the channel with average power constraint. They lead to the following Theorem:

Theorem 4.1 *The capacity of a free-space optical intensity channel with average power constraint is:*

$$C = \log \frac{P_{\mathrm{avg}}}{\sigma} + \frac{1}{2} \log \frac{e}{2\pi} + o(1), \tag{4.10}$$

where the $o(1)$ term tends to 0 as the average power tends to infinity.

Proof: We have to verify (4.10) for the two upper bounds. First, we analyze the upper bound (4.6):

$$\begin{aligned} &\lim_{P_{\mathrm{avg}} \to \infty} \left\{ C_{\mathrm{U,FT}} - \left(\log P_{\mathrm{avg}} + 1 - \frac{1}{2} \log\left(2\pi e \sigma^2\right)\right)\right\} \\ &= \lim_{P_{\mathrm{avg}} \to \infty} \left\{ \frac{2(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})}{P_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}} \right. \\ &\quad + \log\left(\frac{P_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}}{2P_{\mathrm{avg}}}\right) - 1 \\ &\quad \left. + \frac{2\sigma^2}{\left(P_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}\right)^2} + \frac{1}{P_{\mathrm{avg}}} \right\} \\ &= 1 + \log 1 - 1 = 0. \end{aligned} \tag{4.11}$$

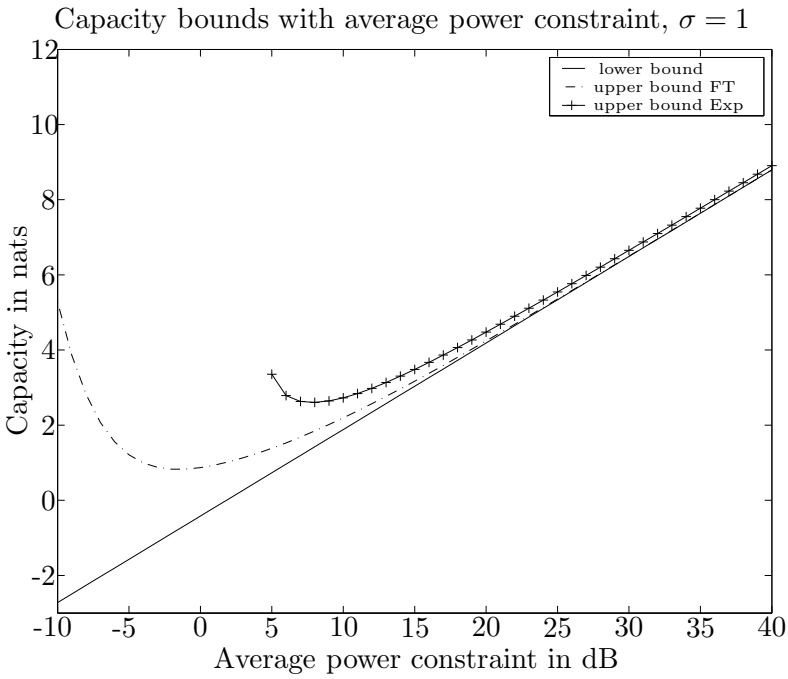

Figure 4.1: Results from Section 4.2. Bounds for the capacity of a free-space optical intensity channel with average power constraint.

Next, we examine (4.8) at high average power:

$$\begin{aligned}
&\lim_{P_{\text{avg}}\to\infty}\left\{C_{\text{U,Exp}} - \left(\log P_{\text{avg}} + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\} \\
&= \log\left(\frac{\sigma}{\sqrt{2\pi}P_{\text{avg}}}e^{-\frac{(\log P_{\text{avg}})^2}{2}} + 1 + \frac{\sigma\log P_{\text{avg}}}{P_{\text{avg}}}\right) - \log\left(1 - \frac{1}{\log P_{\text{avg}}}\right) \\
&\quad + \frac{\sigma\log P_{\text{avg}}}{\sqrt{2\pi}}e^{-\frac{(\log P_{\text{avg}})^2}{2}} \\
&= \log 1 - \log 1 + 0 = 0. \qquad (4.12)
\end{aligned}$$

Comparing these two results with (4.4) leads to (4.10). □

Figure 4.1 shows the lower bound (4.4) and the upper bounds computed with the density of a Fisher-Tippett distribution (4.6) and the density of an exponential distribution(4.8) at the output, respectively.

4.3 Results with Peak Power Constraint only

Contrary to the previous section we are now going to analyse the channel without average power constraint, but with peak power constraint, i.e. the input must lie in the interval $[0, A]$. For this channel model we compute a lower bound in Section 4.3.1 using (2.33). In Section 4.3.2 we apply the dual expression for the capacity (2.17) to obtain an upper bound. These two bounds are asymptotically tight.

4.3.1 Lower Bound

We want to compute a lower bound for the capacity of the channel considering only the peak constraint. As in Section 4.2.1 we use Inequality (4.1):

$$C \geq h(X) - h(Y|X),$$

where again any probability density at the input leads to a lower bound. We choose the density of the uniform distribution between 0 and the peak A:

$$Q(x) = \frac{1}{A}, \quad 0 \leq x \leq A, \tag{4.13}$$

because from Corollary 2.5 we know that this density maximizes the differential entropy over all the densities over the interval $[0, A]$. The differential entropy of the input is then:

$$h(X) = \log(A). \tag{4.14}$$

This gives the following lower bound for the capacity of the channel subject to only a peak power constraint:

$$C \geq C_L = \log(A) - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{4.15}$$

4.3.2 Upper Bound

As in Section 4.2.2 we will use (2.17), where any choice of output density yields an upper bound for the capacity. We will choose:

$$R(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y < -\delta, \\ \frac{P^*}{A+2\delta}, & -\delta \leq y \leq A+\delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y > A+\delta, \end{cases} \tag{4.16}$$

where δ cannot be negative and P^* is the probability that $y \in [-\delta, A+\delta]$:

$$\begin{aligned} P^* &= 1 - \int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \mathrm{d}y - \int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \mathrm{d}y \\ &= \mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right). \end{aligned} \tag{4.17}$$

Using this output density in (2.17) we get the following upper bound for the capacity:

$$\begin{aligned} C \leq\ & \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma} e^{-\frac{(\log A)^2}{2\sigma^2}} \\ & + \log\left(A + 2\log A\right) - \log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right) - \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right) \\ & - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \end{aligned} \tag{4.18}$$

for sufficiently large peak:

$$A \geq 1. \tag{4.19}$$

For a derivation see Appendix A.3.

The choice of the output density (4.16) is not totally random, but actually was suggested to us from the computation of an alternative upper bound. This other bound leads to the same result at high peak power, but is mathematically more difficult. However for the interested reader the alternative upper bound is included in Appendix B.1.

4.3.3 Capacity Results

We want to compare now the bounds obtained in the previous two sections at high peak power.

Theorem 4.2 *The capacity of the free-space optical intensity channel considering only the peak power constraint A is:*

$$C = \log\frac{A}{\sigma} + \frac{1}{2}\log\frac{1}{2\pi e} + o(1), \tag{4.20}$$

where the o(1) term tends to 0 as the peak power constraint tends to infinity.

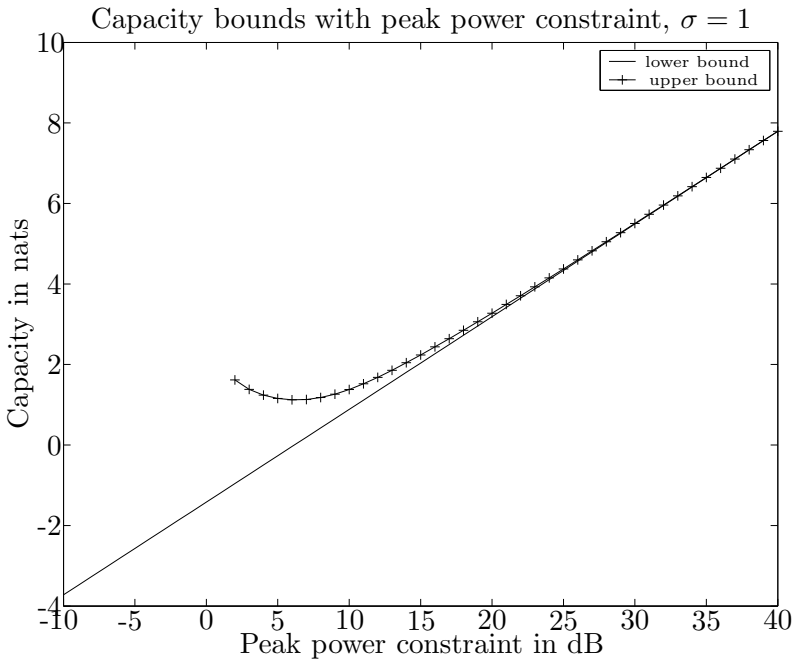

Figure 4.2: Results from Section 4.3. Bounds for the capacity of a free-space optical intensity channel with peak power constraint.

Proof: We analyze the upper bound (4.18) for high peak power:

$$\begin{aligned}
&\lim_{A\to\infty}\left\{C_U - \left(\log A - \log\left(2\pi e\sigma^2\right)\right)\right\} \\
&= \lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}}\right. \\
&\quad \left. + \log\left(1+\frac{2\log A}{A}\right) - \log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right) - \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right)\right\} \\
&= 0 \qquad (4.21)
\end{aligned}$$

Comparing this result with (4.15) we get (4.20). □

In Figure 4.2 the upper bound (4.18) and the lower bound (4.15) are plotted.

4.4 Results with Average Power and Peak Power Constraints

In this section we are going to analyze the capacity of the free-space optical intensity channel with average power and peak power constraints. We will use the same techniques as in the previous sections and the computations will yield again a lower bound and an upper bound which are asymptotically tight at high peak power and fixed ratio between the average power and the peak power. Unfortunately, the results cannot be expressed in closed-form. Instead we have an expression for the capacity with a parameter and an additional condition on this parameter depending on the ratio between the average power and the peak power. The asymptotic results for the capacity of the channels respecting only the average power constraint or only the peak power constraint can also be derived from this result by relaxing one of the two power constraints.

Additionally, we will win an intuition for which ratios between the average power and the peak power constraints both constraints have to be considered and when the error made by relaxing one of the constraints is very small.

4.4.1 Lower Bound

To compute a lower bound for the capacity of the channel with average and peak power constraint we use again the approximation (2.33):

$$C \geq h(X) - h(Y|X).$$

To maximize $h(X)$ and in order to obtain the highest lower bound we use the probability density considering both average power and peak power constraints which maximizes differential entropy. This is the density corresponding to the truncated exponential distribution with average P_{avg} and peak A (see Theorem 2.2 and Corollary 2.3):

$$Q(x) = \frac{\lambda}{1 - e^{-\lambda A}} e^{-\lambda x}, \quad 0 \leq x \leq A, \tag{4.22}$$

where:

$$P_{\text{avg}} = \frac{1}{\lambda} - A \frac{e^{-\lambda A}}{1 - e^{-\lambda A}}. \tag{4.23}$$

Unfortunately, we cannot extract λ from (4.23) and we have to continue the computations with (4.22) and the additional condition (4.23).

The chosen input density leads us to the following lower bound for the differential entropy at the input:

$$\begin{aligned} h(X) &= \int_0^A \frac{\lambda}{1-e^{-\lambda A}} e^{-\lambda x} \log\left(\frac{1-e^{-\lambda A}}{\lambda} e^{\lambda x}\right) \mathrm{d}x \\ &= \log\left(\frac{1-e^{-\lambda A}}{\lambda}\right) + \lambda P_{\text{avg}}, \end{aligned} \tag{4.24}$$

where λ is the solution of (4.23).

Setting the parameter

$$\lambda = \frac{k}{A}, \tag{4.25}$$

we get a lower bound for the capacity with average power constraint P_{avg} and peak power constraint A:

$$C \geq \log(A) + \log\left(\frac{1-e^{-k}}{k}\right) + k\frac{P_{\text{avg}}}{A} - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \tag{4.26}$$

with the condition on k:

$$P_{\text{avg}} = \frac{A}{k} - A\frac{e^{-k}}{1-e^{-k}}. \tag{4.27}$$

4.4.2 Upper Bound

To compute an upper bound for the capacity we use the output density:

$$R(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y < -\delta, \\ \frac{P^*\lambda}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}} e^{-\lambda y}, & -\delta \leq y \leq A+\delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y > A+\delta, \end{cases} \tag{4.28}$$

where δ again may not be negative and P^* is the probability that $y \in [-\delta, A+\delta]$:

$$\begin{aligned} P^* &= 1 - \int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \mathrm{d}y - \int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \mathrm{d}y \\ &= \mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right). \end{aligned} \tag{4.29}$$

Then we apply the density to (2.17) in order to get the following upper bound for the capacity of the channel with average power and peak power

constraints:

$$\begin{aligned} C \leq C_{\mathrm{U}} &= \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}} \\ &+\log\left(\frac{e^{\frac{k\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{k}\right)+\log(A) \\ &-\log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right)-\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right)+k\frac{P_{\mathrm{avg}}}{A} \\ &+\frac{k\sigma}{\sqrt{2\pi}A}\left(e^{-\frac{(\log A)^2}{2\sigma^2}}-e^{-\frac{(A+\log A)^2}{2\sigma^2}}\right) \\ &-\frac{1}{2}\log\left(2\pi e\sigma^2\right) \end{aligned} \tag{4.30}$$

under the additional condition:

$$A \geq 1. \tag{4.31}$$

In (4.30) k is a parameter and can have any nonnegative value. For a derivation see Appendix A.4

Again the choice of the output density is not totally random but was suggested to us by the computation of another upper bound for the capacity. This alternative upper bound can be found in B.2.

4.4.3 Capacity Results

First, we analyze the upper bound (4.30) for high average and peak power. We fix the ratio between the average power and the peak power:

$$\alpha = \frac{P_{\mathrm{avg}}}{A}, \tag{4.32}$$

and let in our analysis the peak power tend to infinity.

Lemma 4.3 *The free-space optical intensity channel with average power constraint P_{avg} and peak power constraint A is upper bounded for fixed ratio $\alpha = \frac{P_{\mathrm{avg}}}{A}$ by:*

$$C_{\mathrm{U}} = \log A + \log\left(\frac{1-e^{-k}}{k}\right) + k\alpha - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1),\ \forall k \geq 0, \tag{4.33}$$

where the $o(1)$ term tends to 0 as the peak power A tends to infinity. The bound is optimized when k fulfills:

$$\alpha = \frac{1}{k} - \frac{e^{-k}}{1-e^{-k}}\,. \tag{4.34}$$

Proof: First, we have to introduce the fixed ratio α in (4.30):

$$\begin{aligned} C \leq C_{\mathrm{U}} &= \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}} \\ &\quad + \log\left(\frac{e^{\frac{k\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{k}\right)+\log(A) \\ &\quad - \log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right)-\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right)+k\alpha \\ &\quad + \frac{k\sigma}{\sqrt{2\pi}A}\left(e^{-\frac{(\log A)^2}{2\sigma^2}}-e^{-\frac{(A+\log A)^2}{2\sigma^2}}\right) \\ &\quad - \frac{1}{2}\log\left(2\pi e\sigma^2\right) \end{aligned} \tag{4.35}$$

and then we have to verify (4.33):

$$\begin{aligned} &\lim_{A\to\infty}\left\{C_{\mathrm{U}}-\left(\log A+\log\frac{1-e^{-k}}{k}+k\alpha-\frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\} \\ &= \lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}}\right. \\ &\quad + \log\left(\frac{e^{\frac{k\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{1-e^{-k}}\right)-\log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right)-\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right) \\ &\quad \left.+\frac{k\sigma}{\sqrt{2\pi}A}\left(e^{-\frac{(\log A)^2}{2\sigma^2}}-e^{-\frac{(A+\log A)^2}{2\sigma^2}}\right)\right\} \\ &= 0. \end{aligned} \tag{4.36}$$

We want to optimize the upper bound for the capacity at high power over k, i.e. we take the derivative of (4.33) and set it equal to 0:

$$\frac{\mathrm{d}C_{\mathrm{U}}}{\mathrm{d}k} = -\frac{1}{k}+\frac{e^{-k}}{1-e^{-k}}+\alpha = 0, \tag{4.37}$$

which is the same condition on k as (4.34). With the second derivative it can be verified that this expression yields a minimum for (4.33). □

Next, we introduce the ratio α in the lower bound (4.26) as well as in its condition (4.27):

$$C \geq C_{\mathrm{L}} = \log A+\log\left(\frac{1-e^{-k}}{k}\right)+k\alpha-\frac{1}{2}\log\left(2\pi e\sigma^2\right), \tag{4.38}$$

$$\alpha = \frac{1}{k}-\frac{e^{-k}}{1-e^{-k}}. \tag{4.39}$$

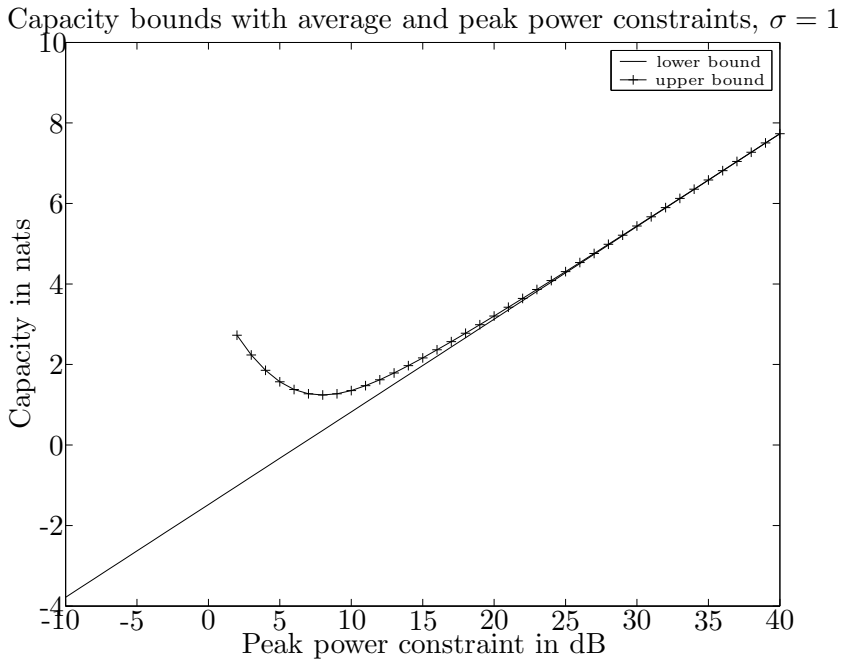

Figure 4.3: Results from Section 4.4. Bounds for the capacity of a free-space optical intensity channel with average and peak power constraints, where $\alpha = 0.4$.

Comparing the lower bound (4.38) and the upper bound from Lemma 4.3 we see that the bounds are tight at high peak power. This leads us to the following theorem:

Theorem 4.4 *The capacity of the free-space optical intensity channel with average optical power constraint P_{avg} and peak optical power constraint A for fixed ratio $\alpha = \frac{P_{\text{avg}}}{A} \leq \frac{1}{2}$ is:*

$$C = \log A + \log\left(\frac{1 - e^{-k}}{k}\right) + k\alpha - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1), \tag{4.40}$$

where the $o(1)$ term tends to 0 as the peak A tends to infinity and k has to fulfill:

$$\alpha = \frac{1}{k} - \frac{e^{-k}}{1 - e^{-k}}. \tag{4.41}$$

In Figure 4.3 the upper bound (4.30) and the lower bound (4.26) are plotted.

Comparison to the Average Power Bound

First, we want to derive the capacity of the channel with only average power constraint at high average power from (4.40). Having only an average power constraint means that the ratio $\alpha = \frac{P_{\text{avg}}}{A}$ becomes 0 because the peak power constraint is infinite. We choose $k = \frac{1}{\alpha}$ and verify that condition (4.41) is fulfilled:

$$\lim_{\alpha\to 0}\left\{\alpha - \frac{1}{k} + \frac{e^{-k}}{1-e^{-k}}\bigg|_{k=\frac{1}{\alpha}}\right\} = \lim_{\alpha\to 0}\left\{\alpha - \alpha + \frac{e^{-\frac{1}{\alpha}}}{1-e^{-\frac{1}{\alpha}}}\right\} = 0. \tag{4.42}$$

Then, we study the influence of the choice α tending towards 0 and $k = \frac{1}{\alpha}$ on the channel capacity:

$$\begin{aligned}
\lim_{\alpha\to 0} C &= \lim_{\alpha\to 0}\left\{\log A + \log\left(\frac{1-e^{-k}}{k}\right) + \alpha k - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1)\bigg|_{k=\frac{1}{\alpha}}\right\} \\
&= \lim_{\alpha\to 0}\left\{\log A + \log\left(\alpha\left(1-e^{-\frac{1}{\alpha}}\right)\right) + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1)\right\} \\
&= \lim_{\alpha\to 0}\left\{\log P_{\text{avg}} + \log\left(1-e^{-\frac{1}{\alpha}}\right) + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1)\right\} \\
&= \log P_{\text{avg}} + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1),
\end{aligned} \tag{4.43}$$

which is exactly the same expression as the asymptotic capacity we have found in (4.10).

Next, we want to determine in which cases the relaxation of the peak power constraint leads only to a small error in the computation of the asymptotic capacity. Therefore, we have to examine the difference between the expressions for the capacity (4.10) and (4.40) at high power.

Lemma 4.5 *Define Δ_A as the difference at high average power between the capacities of the channel subject to an average power and a peak power constraint and the channel with relaxed peak power constraint. Then:*

$$\Delta_A = 1 - \log\frac{1-e^{-k}}{\alpha k} - \alpha k, \tag{4.44}$$

where k must fulfill Condition (4.34).

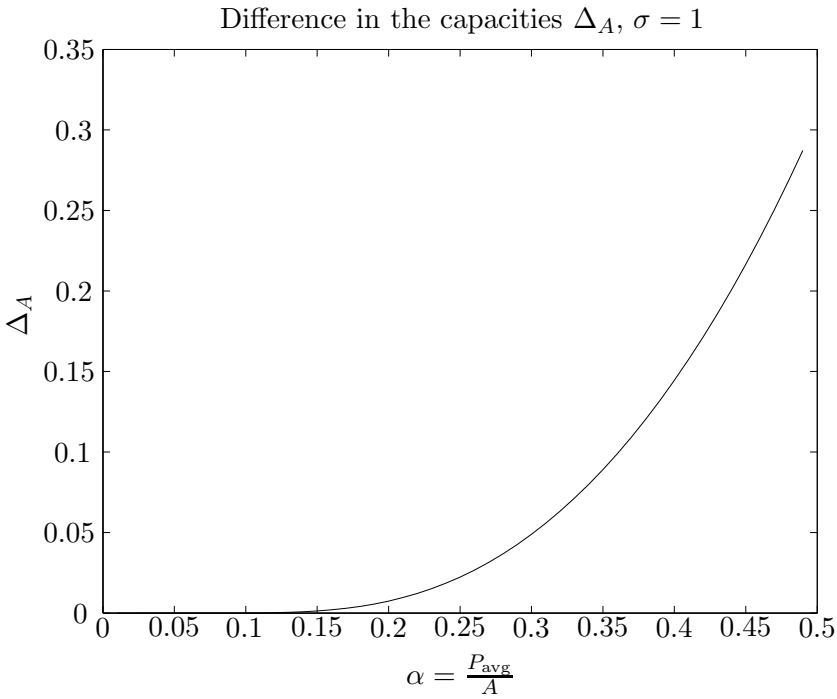

Figure 4.4: Difference of the capacities at high average power of the average power and peak power constrained channel and the only average power constrained channel.

Proof:

$$\begin{aligned}
\Delta_A &= \log P_{\text{avg}} + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right) \\
&\quad - \left(\log A + \log\frac{1-e^{-k}}{k} + \alpha k - \frac{1}{2}\log\left(2\pi e\sigma^2\right)\right) \\
&= \log P_{\text{avg}} - \log\frac{P_{\text{avg}}}{\alpha} + 1 - \log\frac{1-e^{-k}}{k} - \alpha k \\
&= 1 - \log\frac{1-e^{-k}}{k\alpha} - \alpha k. \qquad (4.45)
\end{aligned}$$

□

Figure 4.4 shows the difference Δ_A between the capacities of the average power and peak power constrained channel and the channel with only average power constraint at high average power. We see that Δ_A becomes very small for ratios $\alpha \leq 0.2$.

Comparison to the Peak Power Bound

Now we want to derive the channel capacity with only peak power constraint from (4.40). We know from Section 3.3.3 that the average-to-peak power ratio $\alpha = \frac{1}{2}$. Choosing $k = 0$ we can verify condition (4.41):

$$\begin{aligned}\alpha &= \lim_{k\to 0}\left\{\frac{1}{k} - \frac{e^{-k}}{1-e^{-k}}\right\} = \lim_{k\to 0}\left\{\frac{1-e^{-k}-ke^{-k}}{k\left(1-e^{-k}\right)}\right\} \\ &= \lim_{k\to 0}\left\{\frac{ke^{-k}}{1-e^{-k}+ke^{-k}}\right\} = \lim_{k\to 0}\left\{\frac{e^{-k}-ke^{-k}}{2e^{-k}-ke^{-k}}\right\} = \frac{1}{2}, \end{aligned} \tag{4.46}$$

where in the second and third equality we applied Bernoulli-De l'Hôpital. The channel capacity with only peak power constraint becomes then:

$$\begin{aligned}C &= \lim_{k\to 0}\left\{\log A + \log\left(\frac{1-e^{-k}}{k}\right) + k\alpha - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1)\Big|_{\alpha=\frac{1}{2}}\right\} \\ &= \lim_{k\to 0}\left\{\log A + \log\left(\frac{e^{-k}}{1}\right) - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1)\right\} \\ &= \log A - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1), \end{aligned} \tag{4.47}$$

where again the second equality follows from Bernoulli-De l'Hôpital's rule. This is exactly the same expression at high peak power as we have found in (4.20). With this result we generalize Theorem 4.4 also for $\alpha \geq \frac{1}{2}$.

Corollary 4.6 *The capacity of the optical free-space intensity channel with average power constraint P_{avg} and peak power constraint A for fixed ratio $\alpha \geq \frac{1}{2}$ is:*

$$C = \log\frac{A}{\sigma} + \frac{1}{2}\log\frac{1}{2\pi e} + o(1), \tag{4.48}$$

where the o(1) term tends to 0 as the peak power constraint A tends to infinity.

Proof: From Sections 3.3.2 and 3.3.3 we know that for $\alpha \geq \frac{1}{2}$ the channel behaves as if there was no average power constraint at all. Therefore the capacity is equal to the capacity of the channel with only peak power constraint. □

Analog to the previous section we want to have a closer look at the error which is made in the asymptotic computation of the capacity when

the average power constraint is not considered, but only the peak power constraint.

Lemma 4.7 *Define Δ_P as the difference at high peak power between the capacities of the channel subject to an average power and a peak power constraint and the channel with relaxed average power constraint. Then*

$$\Delta_P = \log \frac{k}{1-e^{-k}} - \alpha k, \tag{4.49}$$

where k fulfills the Condition (4.34).

Proof:

$$\begin{aligned} \Delta_P &= \log A - \frac{1}{2}\log\left(2\pi e\sigma^2\right) \\ &\quad - \left(\log A + \log\frac{1-e^{-k}}{k} + \alpha k - \frac{1}{2}\log\left(2\pi e\sigma^2\right)\right) \\ &= \log\frac{k}{1-e^{-k}} - \alpha k. \end{aligned} \tag{4.50}$$

□

Figure 4.5 shows the difference Δ_P. We see that the error we make by not considering the average power constraint is very small when the ratio $\alpha \geq 0.4$.

Combining Figures 4.4 and 4.5 we see that at high power we can neglect either the average power constraint or the peak power constraint to compute the channel capacity with an error of less than 0.15 nats.

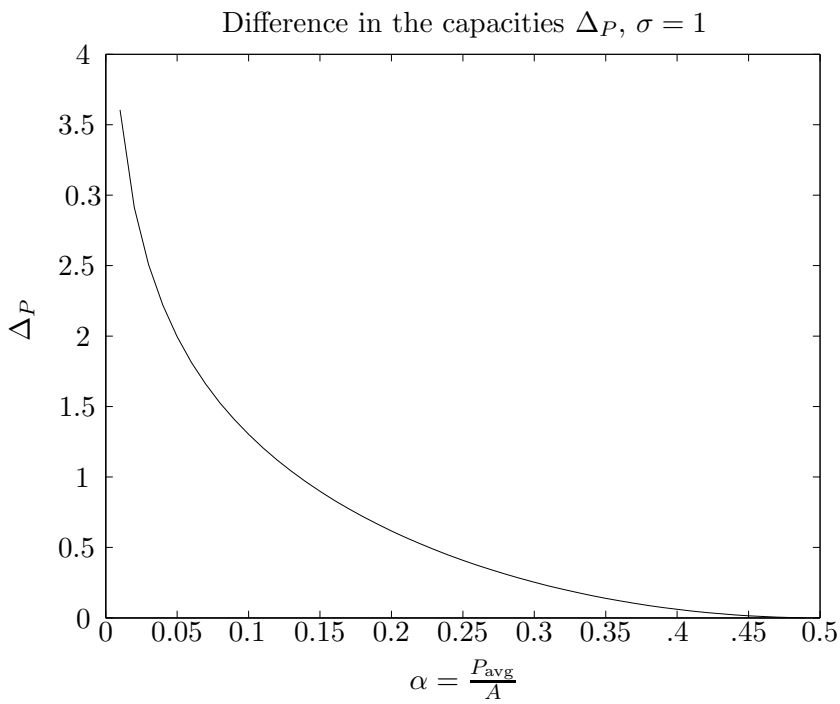

Figure 4.5: Difference of the capacities at high peak power of the average power and peak power constrained channel and the only peak power constrained channel.

Chapter 5

The Channel Capacity at Low Power

The bounds found in Chapter 4 are quite bad for low power and in some cases even not valid. Therefore, we have to compute new lower and upper bounds for the optical free-space intensity channel that are good at low optical power (average or peak).

As in Chapter 4 the same three different cases of the channel model are considered. For the peak power constrained channels (with or without additional average power constraint) we are going to present upper and lower bounds which both tend to 0 for low power, whereas for the capacity of the average power constrained channel we could not find any upper bound which tends to 0.

The bounds found for the peak power constrained channel can directly be deduced from the bounds respecting both constraints by relaxing the average power constraint. Similarly, we can derive the lower bound of the only average power constrained channel by relaxing the peak power constraint.

5.1 Introduction

In this chapter we want to present upper and lower bounds for the channel capacity which are good for low power. This means that they tend to 0 as the peak or the average power constraints tend to 0. For the lower bounds the power entropy inequality from Theorem 2.6 is used. To compute the upper bounds we will again make use of (2.17) as we did to find the upper bounds

at high power. But this time we will consider different output densities, so as to obtain good bounds at low power.

Again as in the previous chapter first a simplified channel model is analyzed, i.e., only the average power constraint is considered and not the peak power. Then, in the next section the average power constraint is relaxed and instead a peak power constraint is considered. Finally, in Section 5.4 both constraints are taken into account.

5.2 Results with Average Power Constraint only

From Figure 4.1 it can be seen that the upper and lower bounds for the capacity of the only average power constrained channel found in section 4.2 are very bad for low average power. Therefore in this section a lower bound is computed using the stronger version of the entropy power inequality (2.31). This will yield a bound which tends to 0 for low average power unlike the lower bound (4.4) which tends to $-\infty$. Then an upper bound is found with the same method as for high power, but by choosing different parameters for the density of the Fisher-Tippett distribution at the output. The received bound does not tend to 0 for low average, but to a small fixed value.

5.2.1 Lower Bound

Again we will use the entropy power inequality to obtain a lower bound, but this time the stronger version (2.31). Using the input density in the interval $[0, \infty]$ with average constraint P_{avg} which maximizes differential entropy:

$$Q(x) = \frac{1}{P_{\text{avg}}} e^{-\frac{x}{P_{\text{avg}}}}, \tag{5.1}$$

we obtain the following lower bound for the differential entropy at the output:

$$\begin{aligned} h(Y) &\geq \frac{1}{2} \log \left(e^{2h(X)} + e^{2h(Z)} \right) \\ &= \frac{1}{2} \log \left(P_{\text{avg}}^2 e^2 + 2\pi e \sigma^2 \right). \end{aligned} \tag{5.2}$$

So we have found a lower bound at low power that tends to 0 when the power tends to 0:

$$C = h(Y) - h(Y|X) \geq C_{\text{L}} = \frac{1}{2} \log \left(1 + \frac{P_{\text{avg}}^2 e}{2\pi\sigma^2} \right). \tag{5.3}$$

5.2.2 Upper Bound

The upper bound for low average power is computed with the same expression for the capacity (2.17) and using again the density of a Fisher-Tippett distribution at the output as in section 4.2.2. Only the parameters of the density will be chosen differently.

So using the density of the Fisher-Tippett distribution with parameters a and b:

$$R(y) = \frac{e^{\frac{a-y}{b}} - e^{\frac{a-y}{b}}}{b}, \tag{5.4}$$

gives an upper bound for the capacity (see A.1):

$$C \leq C_{\mathrm{U}} = \frac{-a}{b} + \log b + \frac{P_{\mathrm{avg}}}{b} + e^{\frac{a}{b}+\frac{\sigma^2}{2b^2}} - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{5.5}$$

In order to obtain a low upper bound at low average power the parameters are chosen:

$$a = \frac{-\sigma^2}{2b}, \tag{5.6}$$

$$b = \frac{P_{\mathrm{avg}} + \sqrt{P_{\mathrm{avg}}^2 + 4\sigma^2}}{2}. \tag{5.7}$$

This yields an upper bound for the capacity of the only average power constrained channel:

$$\begin{aligned} C_{\mathrm{U}} &= \frac{2P_{\mathrm{avg}}}{P_{\mathrm{avg}} + \sqrt{P_{\mathrm{avg}}^2 + 4\sigma^2}} + \log\left(\frac{\mathrm{P}_{\mathrm{avg}} + \sqrt{P_{\mathrm{avg}}^2 + 4\sigma^2}}{2}\right) \\ &\quad + \frac{2\sigma^2}{\left(P_{\mathrm{avg}} + \sqrt{P_{\mathrm{avg}}^2 + 4\sigma^2}\right)^2} + 1 - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{5.8}$$

5.2.3 Capacity Results

We want to compare the obtained upper and lower bounds for low average power. The behaviour of the lower bound for small average is straight forward, whereas the upper bound for low average power requires some closer examination.

Lemma 5.1 *The capacity of the average power constrained channel is upper bounded by:*

$$C_{\mathrm{U}} = \log\left(\frac{P_{\mathrm{avg}}e}{2\sqrt{2\pi}\sigma} + \frac{e}{\sqrt{2\pi}}\right) + f(P_{\mathrm{avg}}), \tag{5.9}$$

where $f(P_{\mathrm{avg}})$ tends to 0 if P_{avg} tends to 0.

Proof: We use the upper bound (5.8):

$$\begin{aligned}
&\lim_{P_{\text{avg}}\to 0}\left\{C_{\text{U}} - \log\left(\frac{P_{\text{avg}}e}{2\sqrt{2\pi}\sigma} + \frac{e}{\sqrt{2\pi}}\right)\right\} \\
&= \lim_{P_{\text{avg}}\to 0}\left\{\frac{2P_{\text{avg}}}{P_{\text{avg}} + \sqrt{P_{\text{avg}}^2 + 4\sigma^2}} + \log\left(\frac{P_{\text{avg}}}{P_{\text{avg}} + 2\sigma} + \frac{\sqrt{P_{\text{avg}}^2 + 4\sigma^2}}{P_{\text{avg}} + 2\sigma}\right)\right. \\
&\quad \left. + \frac{2\sigma^2}{\left(P_{\text{avg}} + \sqrt{P_{\text{avg}}^2 + 4\sigma^2}\right)^2} - \frac{1}{2}\right\} \\
&= 0 + \log 1 + \frac{1}{2} - \frac{1}{2} = 0. \qquad (5.10)
\end{aligned}$$

□

So the upper bound for the capacity tends to $\log\left(\frac{e}{\sqrt{2\pi}}\right) = 0.0811$ nats when the average power tends to 0.

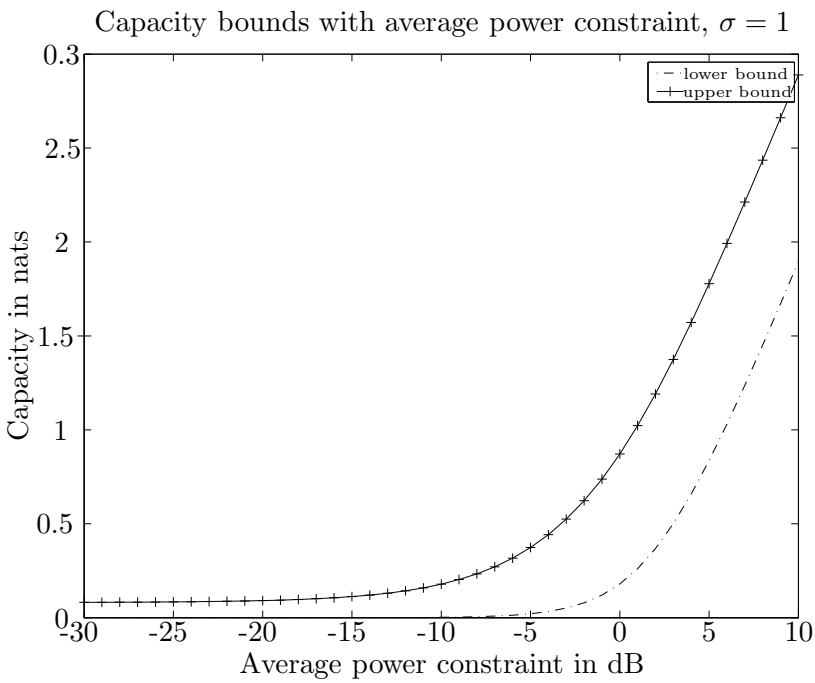

Figure 5.1: Results from section 5.2. Bounds for the capacity of a free-space optical intensity channel with average power constraint at low power.

Figure 5.1 shows the upper bound (5.8) and the lower bound (5.3) for low power. We can see that the bounds do not tend towards each other for small average power, especially the upper bound is bounded away from 0.

However, no upper bound could be found for the only average power constrained channel which tends to 0 at low power. In section 5.4 we will see as well that no better bound than (5.8) can be derived from the upper bound found for the capacity of the channel respecting both an average power and a peak power constraint. Most probably the problem is that the output density which is capacity achieving tends to a Gaussian density for a small average power constraint. But we don't know how to compute the expression (2.17) with a Gaussian density at the output, as we cannot upper bound the second moment of the capacity achieving input density without peak constraint.

5.3 Results with Peak Power Constraint only

In this section we derive an upper bound and a lower bound for the capacity of the channel with only peak power constraint. These bounds are good at low power, i.e. both tend to 0 as the peak power tends to 0. In Section 5.3.3 we discuss and compare the two obtained bounds at low power.

5.3.1 Lower Bound

The lower bound in Section 4.3 tends to $-\infty$ for small peaks. Therefore, we are looking for a better lower bound at low power and use the stronger version of the entropy power inequality (2.31). Together with the input density in the interval $[0, A]$ which maximizes differential entropy:

$$Q(x) = \frac{1}{A}, \tag{5.11}$$

we obtain the following bound for the differential output entropy:

$$\begin{aligned} h(Y) &\geq \frac{1}{2}\log\left(e^{2h(X)} + e^{2h(Z)}\right) \\ &= \frac{1}{2}\log\left(A^2 + 2\pi e\sigma^2\right). \end{aligned} \tag{5.12}$$

Thus, we have found a new lower bound which tends to 0 at low power:

$$C = h(Y) - h(Y|X) \geq C_{\mathrm{L}} = \frac{1}{2}\log\left(1 + \frac{A^2}{2\pi e\sigma^2}\right). \tag{5.13}$$

5.3.2 Upper Bound

To compute an upper bound which is good at low power, meaning that it tends to 0 as the power vanishes, we choose the following Gaussian density at the output:

$$R(y) = \frac{1}{\sqrt{2\pi}L} e^{-\frac{(y-m)^2}{2L^2}}. \tag{5.14}$$

Next we apply the chosen output density to compute the first term of (2.17):

$$\begin{aligned}
&\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X)\log(R(y))\,\mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}L e^{\frac{(y-m)^2}{2L^2}}\right)\mathrm{d}y\right] \\
&= \log\left(\sqrt{2\pi}L\right) + \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \frac{(y-m)^2}{2L^2}\,\mathrm{d}y\right] \\
&= \log\left(\sqrt{2\pi}L\right) + \mathrm{E}^*_Q\left[\frac{X^2+\sigma^2-2mX+m^2}{2L^2}\right] \\
&\leq \log\left(\sqrt{2\pi}L\right) + \frac{\sigma^2+\frac{A^2}{2}-mA+m^2}{2L^2},
\end{aligned} \tag{5.15}$$

where the inequality follows from $\mathrm{E}[X^2] \leq A \cdot \mathrm{E}[X]$ and Theorem 3.3. This expression yields an upper bound for the capacity with parameters m and $L \geq 0$:

$$C \leq C_{\mathrm{U}} = \log\left(\sqrt{2\pi}L\right) + \frac{\sigma^2+\frac{A^2}{2}-mA+m^2}{2L^2} - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{5.16}$$

Now we have to find the minimal upper bound. By differentiating (5.16) by m and L and setting the expressions to 0, we receive the following values for m and L:

$$m = \frac{A}{2}, \tag{5.17}$$

$$L^2 = \sigma^2 + \frac{A^2}{4}. \tag{5.18}$$

It can be verified that the second derivatives of (5.16) are positive for these values and therefore we receive a minimum.

This gives an upper bound for the capacity of the peak power constrained channel, which tends to 0 at low power:

$$C \leq \frac{1}{2}\log\left(2\pi\left(\sigma^2+\frac{A^2}{4}\right)\right) + \frac{1}{2} - \frac{1}{2}\log\left(2\pi e\sigma^2\right) = \frac{1}{2}\log\left(1+\frac{A^2}{4\sigma^2}\right). \tag{5.19}$$

5.3.3 Capacity Results

The behaviour of the upper and the lower bound found in the previous two sections for low peak power is straight forward. Figure 5.2 shows the two bounds at low power and it can be seen, that both bounds tend to 0 when the peak power tends to 0.

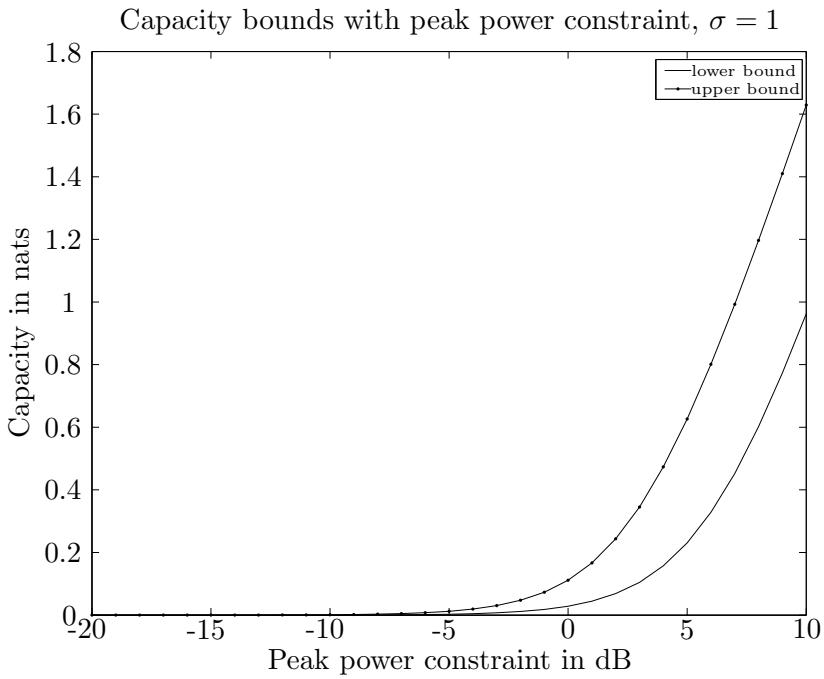

Figure 5.2: Results from Section 5.3. Bounds for the capacity of a free-space optical intensity channel with peak power constraint at low power.

Smith proved in [10] that the capacity achieving input density of an AWGN channel with input in the interval $[0, A]$ is discrete. Especially, he showed that for low power the capacity achieving input density is of the form:

$$Q^*(x) = \frac{1}{2}\delta(x) + \frac{1}{2}\delta(x - A). \tag{5.20}$$

This input density leads to an output density which is a superposition of two Gaussian densities with mean 0 and A, respectively, and variance σ^2. If we use this output density together with the dual expression (2.17) we get the following upper bound for the capacity (see C.1):

$$C \leq \frac{A^2}{4\sigma^2}, \tag{5.21}$$

which is clearly higher than (5.19). The difference in the bounds arises from the fact that we cannot analytically compute (2.17) for the capacity achieving output density, but we have to upper bound some expressions.

This shows that sometimes it is preferable to use an output density which is not as close to the capacity achieving density but leads to a mathematically more tractable result.

5.4 Results with Average Power and Peak Power Constraints

In this section we want to consider both the average and the peak power constraint. We are going to compute an upper and a lower bound, where from the lower bound we can derive the lower bounds of the channels with only one constraint by relaxing the other constraint. In contrast to this, from the upper bound it will only be possible to deduce the upper bound for the capacity of the peak power constrained channel.

The bounds found in this section tend both to 0 as the peak power tends to 0, where the ratio $\alpha = \frac{P_{\text{avg}}}{A}$ is kept constant.

5.4.1 Lower Bound

As in the previous sections we compute the lower bound with the help of the entropy power inequality. We use the differential entropy maximizing input density respecting the average P_{avg} and the peak A found in Section 2.3:

$$Q(x) = \frac{\lambda}{1 - e^{-\lambda A}} e^{-\lambda x}, \tag{5.22}$$

where the parameter λ has to fulfill the following equation:

$$P_{\text{avg}} = \frac{1}{\lambda} - \frac{e^{-\lambda A} A}{1 - e^{-\lambda A}}. \tag{5.23}$$

The maximum differential entropy is:

$$h(X) = \log\left(\frac{1 - e^{-\lambda A}}{\lambda}\right) + \lambda P_{\text{avg}}, \tag{5.24}$$

and the entropy power inequality leads to:

$$\begin{aligned} h(Y) &\geq \frac{1}{2}\log\left(e^{2h(X)} + e^{h(Z)}\right) \\ &= \frac{1}{2}\log\left(e^{2\lambda P_{\text{avg}}}\left(\frac{1 - e^{-\lambda A}}{\lambda}\right)^2 + 2\pi e \sigma^2\right). \end{aligned} \tag{5.25}$$

Hence, we have found a lower bound for the capacity of the channel with average and peak power constraint:

$$C \geq C_{\mathrm{L}} = \frac{1}{2}\log\left(1+\frac{e^{2\lambda P_{\mathrm{avg}}}}{2\pi e\sigma^2}\left(\frac{1-e^{-\lambda A}}{\lambda}\right)^2\right), \tag{5.26}$$

where the condition (5.23) for the parameter λ has to be fulfilled.

5.4.2 Upper Bound

To compute an upper bound we choose the following output density:

$$R(y) = \frac{1}{\sqrt{2\pi}L}e^{-\frac{(y-m)^2}{2L^2}}, \tag{5.27}$$

where L has to be positive. Then, we can compute the first term of (2.17) similarly to Section 5.3.2:

$$\begin{aligned}
&\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X)\log(R(y))\,\mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(y-X)^2}{2\sigma^2}}\log\left(\sqrt{2\pi}Le^{\frac{(y-m)^2}{2L^2}}\right)\mathrm{d}y\right] \\
&= \log\left(\sqrt{2\pi}L\right)+\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(y-X)^2}{2\sigma^2}}\frac{(y-m)^2}{2L^2}\,\mathrm{d}y\right] \\
&= \log\left(\sqrt{2\pi}L\right)+\mathrm{E}_{Q^*}\left[\frac{X^2+\sigma^2-2mX+m^2}{2L^2}\right] \\
&\leq \log\left(\sqrt{2\pi}L\right)+\frac{\sigma^2+P_{\mathrm{avg}}A-2mP_{\mathrm{avg}}+m^2}{2L^2},
\end{aligned} \tag{5.28}$$

where in the inequality we again used that $\mathrm{E}[X^2] \leq A\cdot\mathrm{E}[X]$. To obtain a minimal upper bound for the capacity we have to find the optimal values for the parameters L and m:

$$m = P_{\mathrm{avg}}, \tag{5.29}$$

$$L^2 = \sigma^2 + P_{\mathrm{avg}}\left(A - P_{\mathrm{avg}}\right). \tag{5.30}$$

The values have been found by taking the derivatives of (5.28) and setting them to 0. With the second derivatives it can be verified that we have obtained a minimum for the upper bound (5.28).

This leads us to the following upper bound for the capacity of the channel considering an average and a peak power constraint:

$$C \leq C_{\mathrm{U}} = \frac{1}{2}\log\left(1+P_{\mathrm{avg}}\left(A-P_{\mathrm{avg}}\right)\right). \tag{5.31}$$

5.4.3 Capacity Results

We want to compare the two bounds found in the previous sections for low power. Therefore, we introduce the ratio $\alpha = \frac{P_{\text{avg}}}{A}$ in (5.26) and (5.31). Additionally we replace in the lower bound $\lambda = \frac{k}{A}$:

$$C_{\text{L}} = \frac{1}{2} \log \left(1 + e^{2k\alpha} \left(\frac{1 - e^{-k}}{k} \right)^2 \frac{A^2}{2\pi e \sigma^2} \right). \tag{5.32}$$

Equation (5.23) leads to the condition for the parameter k:

$$\alpha = \frac{1}{k} - \frac{e^{-k}}{1 - e^{-k}}. \tag{5.33}$$

By introducing the ratio α in the upper bound we receive:

$$C_{\text{U}} = \frac{1}{2} \log \left(1 + \alpha(1 - \alpha) A^2 \right). \tag{5.34}$$

Figures 5.3 and 5.4 show the two bounds at low peak power with fixed ratios

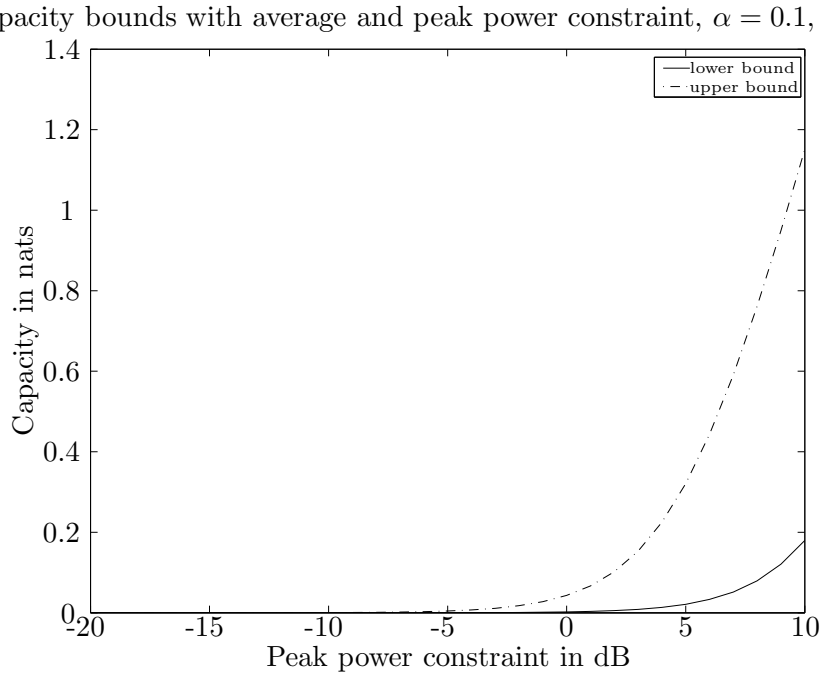

Figure 5.3: Results from Section 5.4. Bounds for the capacity of a free-space optical intensity channel with average power and peak power constraint at low peak power and fixed ratio $\alpha = 0.1$.

$\alpha = 0.1$ and $\alpha = 0.4$, respectively. We notice that the difference between the lower and the upper bound increases faster in Figure 5.3 than in Figure 5.4.

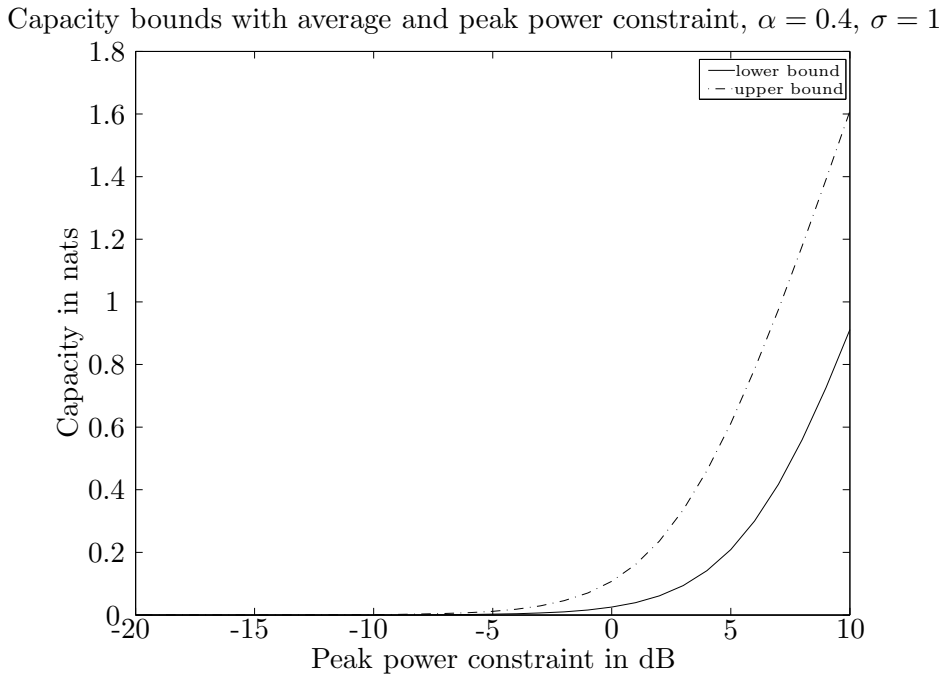

Figure 5.4: Results from Section 5.4. Bounds for the capacity of a free-space optical intensity channel with average power and peak power constraint at low peak power and fixed ratio $\alpha = 0.4$.

Comparison to the Peak Power Bounds

In the obtained lower and upper bounds for the capacity of the channel subject to an average and a peak power constraint (5.32) and (5.34) we can relax the average power constraint to obtain the same bounds as in Section 5.3. Theorem 3.3 tells us that when relaxing the average power constraint we have to choose $\alpha = \frac{1}{2}$.

We examine first the lower bound (5.32). Then the parameter k has to be equal to 0 in order to fulfill (5.33), see Section 4.4.3. With these two values we get the following lower bound for the only peak power constrained channel:

$$\begin{aligned}
\lim_{k\to 0} C_{\mathrm{L}} &= \lim_{k\to 0} \frac{1}{2} \log \left(1 + e^k \left(\frac{1-e^{-k}}{k} \right)^2 \frac{A^2}{2\pi e \sigma^2} \right) \\
&= \lim_{k\to 0} \frac{1}{2} \log \left(1 + \left(\frac{e^{-k}}{1} \right)^2 \frac{A^2}{2\pi e \sigma^2} \right) \\
&= \frac{1}{2} \log \left(1 + \frac{A^2}{2\pi e \sigma^2} \right), \qquad (5.35)
\end{aligned}$$

where in the second equality we applied Bernoulli-De l'Hôpital. This expression is the same as in (5.13).

Next we want to relax the average power constraint in the upper bound. Therefore we insert $\alpha = \frac{1}{2}$ in (5.34):

$$C_{\mathrm{U}} = \frac{1}{2} \log \left(1 + \frac{A^2}{4} \right). \qquad (5.36)$$

Again we can see that we get the same expression as in (5.19).

Comparison to the Average Power Bounds

We can relax as well the peak power constraint in the obtained upper and lower bounds (5.32) and (5.34). Then α has to tend to 0 because the peak is infinite.

First we want to examine the lower bound. To fulfill condition (5.33) k has to be set equal to $\frac{1}{\alpha}$ (see Section 4.4.3). Putting these values into (5.32)

we obtain:

$$\begin{aligned}
\lim_{\alpha\to 0} C_{\mathrm{L}} &= \lim_{\alpha\to 0} \frac{1}{2}\log\left(1+e^2\left(\left(1-e^{-\frac{1}{\alpha}}\right)\alpha\right)^2 \frac{P_{\mathrm{avg}}^2}{2\pi e\sigma^2\alpha^2}\right) \\
&= \lim_{\alpha\to 0} \frac{1}{2}\log\left(1+e^2\left(1-e^{-\frac{1}{\alpha}}\right)^2 \frac{P_{\mathrm{avg}}^2}{2\pi e\sigma^2}\right) \\
&= \frac{1}{2}\log\left(1+\frac{P_{\mathrm{avg}}^2 e}{2\pi\sigma^2}\right), \qquad (5.37)
\end{aligned}$$

which is equal to the previously found lower bound (5.3).

If we introduce $\alpha = 0$ in the upper bound for the capacity of the channel with average and peak power constraint (5.34), then we obtain an infinite bound. This means that this upper bound is only good if a non-neglectable peak power constraint has to be considered.

Chapter 6

Discussion

In the previous chapters we have found and discussed lower and upper bounds for the three different channel models either at low or at high power. Now we want to give an insight on how the bounds behave in all the power regions, as well in the area that does not clearly belong to either low or high power. We will see that actually in this area the difference between lower and upper bounds is largest.

In the second part of this chapter some earlier results on the capacity of the free-space optical intensity channel are presented and compared to our own results. All the previous results consider only the average power constraint and the non-negativity constraint of the input and hold for high optical power. So most of our results are new and cannot be compared to previous results.

6.1 Discussion of the Results of Chapters 4 and 5

In this section we want to give an overview about the results obtained in the previous two chapters. We will show the bounds for low and high power together and estimate the maximal difference between the upper and lower bounds. This difference is maximal in the region which cannot be classified to high or low power. Especially for high power and in the cases with peak constraint as well for low power the difference between the upper and the lower bounds vanish.

First, we analyze the channel subject to only an average power constraint. We can see in Figure 6.1 that the maximal absolute difference

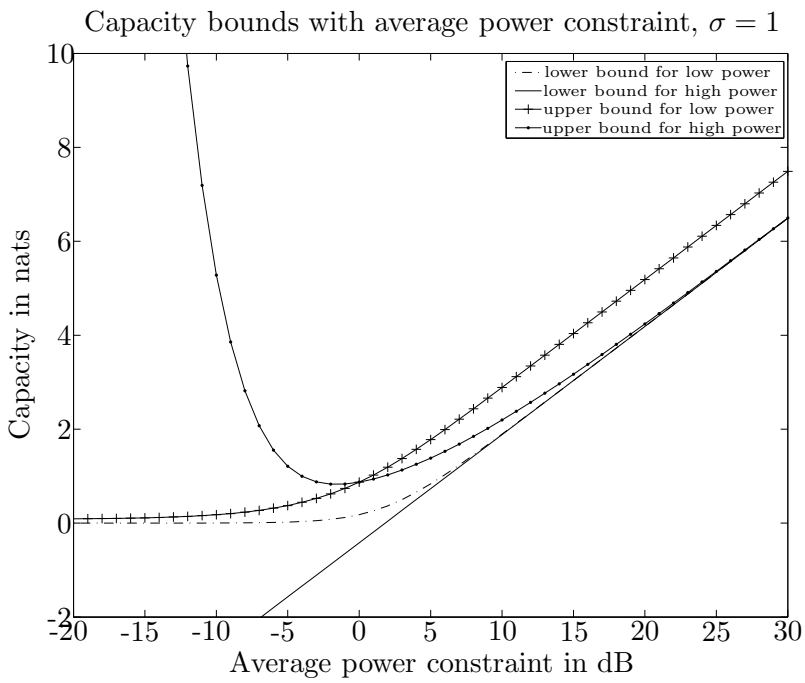

Figure 6.1: Bounds for the channel capacity considering only the average power constraint.

between the upper and lower bound is about 0.8 nats and lies around the 0 dB point. If we examine the difference relative to the true capacity, we see that it is greatest when the average power tends to 0. There the upper bound tends to $\log\left(\frac{e}{\sqrt{2\pi}}\right) = 0.0811$ nats, whereas the lower bound and as well the capacity tend to 0.

Next, we analyze the bounds considering only a peak power constraint. In this case the difference of the upper and the lower bound tends to 0 for low and for high peak power. In particular for low peak power we get an upper bound which tends to 0. From Figure 6.2 we can see that the maximal difference between the best found upper bound and the best found lower bound lies around the 8 dB point and is only 0.5 nats. The asymptotic upper bound for high peak power for an AWGN channel which has input in the interval $[0, A]$ has already been found by Shannon [11]. We now obtained a firm upper bound. Additionally, we can show in Figure 6.2 that the capacity tends quite fast to this bound.

Finally, we consider the capacity of the free-space optical intensity channel subject to both an average and a peak power constraint. Again the

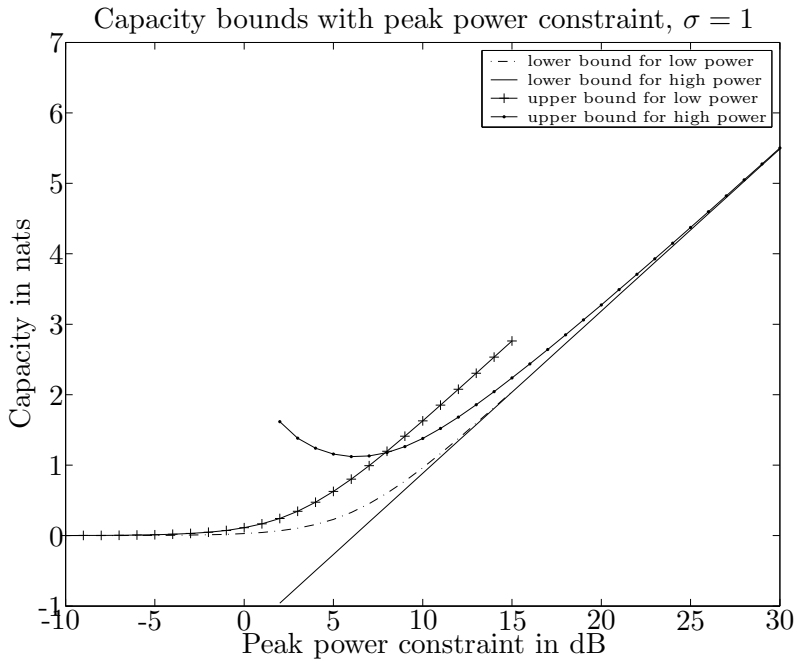

Figure 6.2: Bounds for the channel capacity considering only the peak power constraint.

difference of upper and the lower bounds vanishes for low and high power. Figure 6.3 shows the bounds for the average-to-peak power ratio $\alpha = 0.3$. The maximal difference is situated around the 8 dB point and is about 0.7 nats.

6.1.1 Comparison of the Bounds of the Peak Power Constrained Channel to the Bounds of the Average and Peak Power Constrained Channel

After looking at the three different channel models all by their own, we want to find a relationship between the different bounds. First, we look at the bounds for the peak power constrained channels with and without additional average power constraint. As we have already seen in the previous chapters all the bounds found for the capacity of the channel with peak power constraint can be derived from the capacity of the channel subject to an additional average power constraint by setting $\alpha = \frac{1}{2}$. This holds not only asymptotically! In Figure 6.4 we therefore can see that when the ratio α tends to $\frac{1}{2}$ then the bounds approximate the bounds for the channel

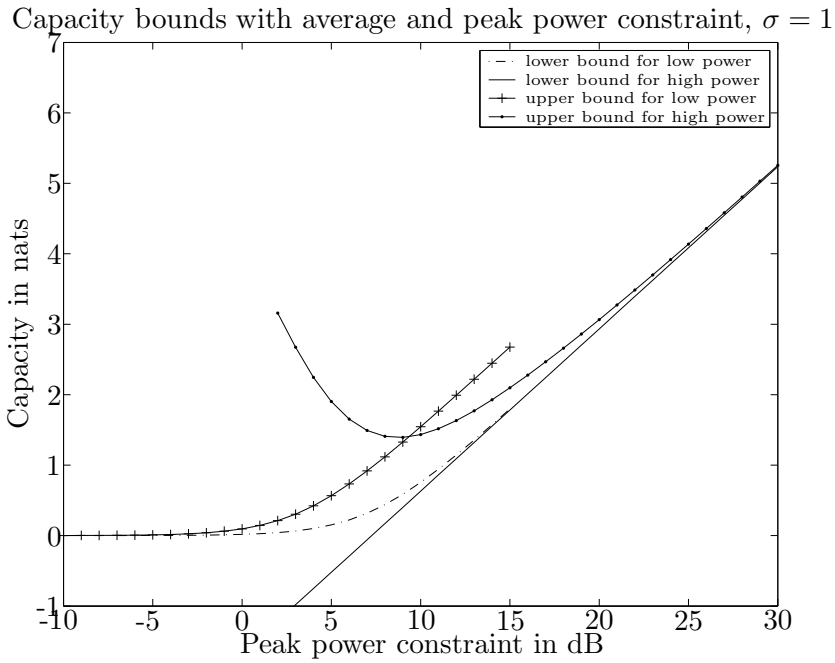

Figure 6.3: Bounds for the channel capacity considering the average and the peak power constraints with fixed ratio $\alpha = 0.3$.

considering only the peak power constraint, which are represented in the last plot.

6.1.2 Comparison of the Bounds of the Average Power Constrained Channel to the Bounds of the Average and Peak Power Constrained Channel

We have already found lower and upper bounds for the capacity of the channel respecting average and peak power constraints with fixed ratio $\alpha = \frac{P_{\text{avg}}}{A}$ in function of the peak power. Now we want to plot the bounds in function of the average power in order to compare graphically the results to the bounds obtained for the only average power constrained channel. Therefore, we have to change the expressions into bounds in function of the average power.

We start with the lower bound which is good at low power (5.32). It can be expressed in terms of P_{avg} and α as:

$$C \geq \frac{1}{2} \log \left(1 + e^{k\alpha} \left(\frac{1 - e^{-k}}{k\alpha} \right)^2 \frac{P_{\text{avg}}^2}{2\pi e \sigma^2} \right), \tag{6.1}$$

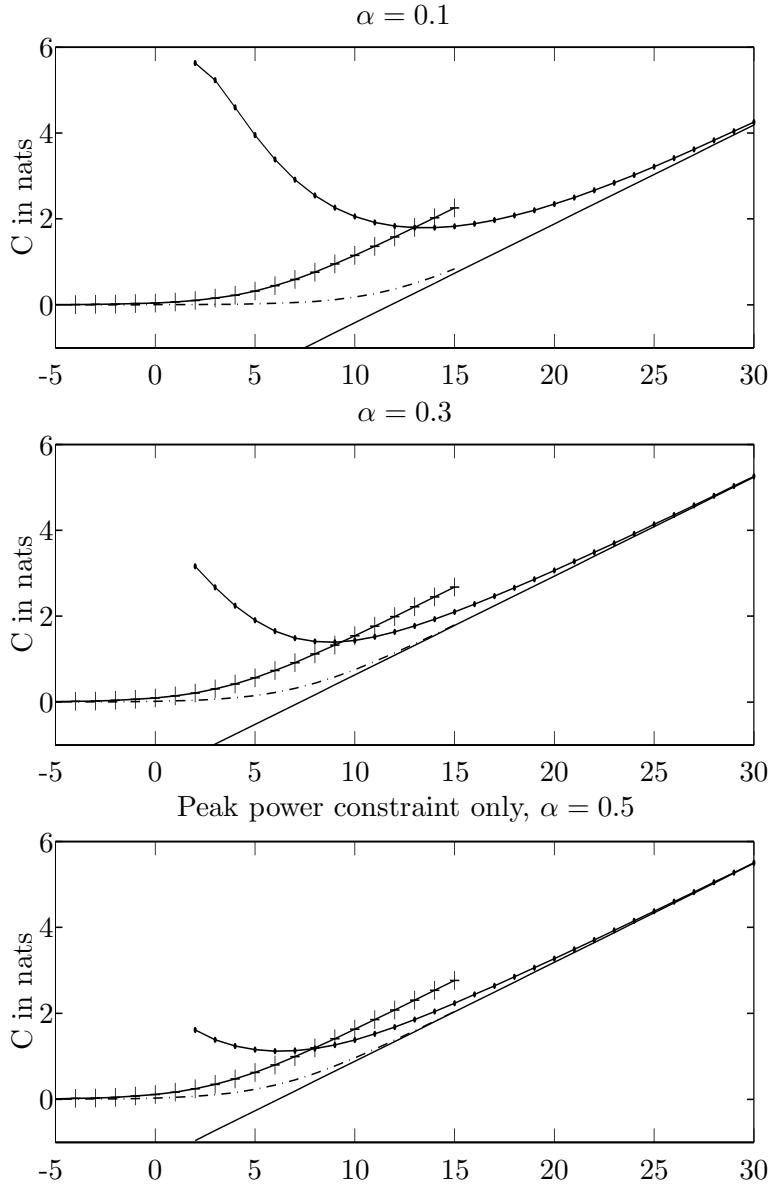

Figure 6.4: Evolution of the bounds for average and peak power constrained channels in function of the peak power with increasing α.

where k has to fulfill the condition:

$$\alpha = \frac{1}{k} - \frac{e^{-k}}{1 - e^{-k}}. \tag{6.2}$$

The lower bound (4.26) expressed as a function of the average power con-

straint gives:

$$C \geq \log P_{\text{avg}} + \log\left(\frac{1-e^{-k}}{k\alpha}\right) + k\alpha - \frac{1}{2}\log\left(2\pi e\sigma^2\right), \tag{6.3}$$

where (6.2) has to be fulfilled.

Then, the upper bounds in function of the average power instead of the peak power are (derived from (5.34)):

$$C \leq \frac{1}{2}\log\left(1 + \frac{(1-\alpha)}{\alpha}P_{\text{avg}}^2\right), \tag{6.4}$$

and (deduced from the upper bound (4.30)):

$$\begin{aligned} C \;\leq\; & \mathcal{Q}\left(\frac{\log P_{\text{avg}} - \log\alpha}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) \\ & + \frac{\log P_{\text{avg}} - \log\alpha}{\sqrt{2\pi}\sigma} e^{-\frac{(\log P_{\text{avg}} - \log\alpha)^2}{2\sigma^2}} + \log P_{\text{avg}} \\ & + \log\left(\frac{e^{\frac{k\alpha(\log P_{\text{avg}} - \log\alpha)}{P_{\text{avg}}}} - e^{-k\left(1+\frac{\alpha(\log P_{\text{avg}} - \log\alpha)}{P_{\text{avg}}}\right)}}{k\alpha}\right) \\ & - \log\left(\mathcal{Q}\left(-\frac{\log P_{\text{avg}} - \log\alpha}{\sigma}\right) - \mathcal{Q}\left(\frac{\log P_{\text{avg}} - \log\alpha}{\sigma}\right)\right) + k\alpha \\ & + \frac{k\alpha\sigma}{\sqrt{2\pi}P_{\text{avg}}}\left(e^{-\frac{(\log P_{\text{avg}} - \log\alpha)^2}{2\sigma^2}} - e^{-\frac{\left(\frac{P_{\text{avg}}}{\alpha} + \log P_{\text{avg}} - \log\alpha\right)^2}{2\sigma^2}}\right) \\ & - \frac{1}{2}\log\left(2\pi e\sigma^2\right), \end{aligned} \tag{6.5}$$

where again (6.2) must hold for k.

We have seen in the previous chapters that from the lower bounds for the capacity of the average and peak power constrained channel the lower bounds for the only average power constrained channel can be derived. This is not possible for the upper bounds. For α tending to 0 the upper bound for low power becomes infinite, whereas the upper bound for high power only asymptotically converges to the capacity bound we have already found for the channel with only average power constraint (4.10). Figure 6.5 shows the evolution of the bounds for the capacity of the channel subject to an average and a peak power constraint in function of the average power for decreasing α. The last plot shows the bounds for the channel with average power constraint only. We can see in the figure that indeed the lower bounds and the upper bound for high power tend to the corresponding bounds which

consider only an average power constraint. For the upper bound at low average power this is not the case as the bound tends to infinity, as we mentioned before.

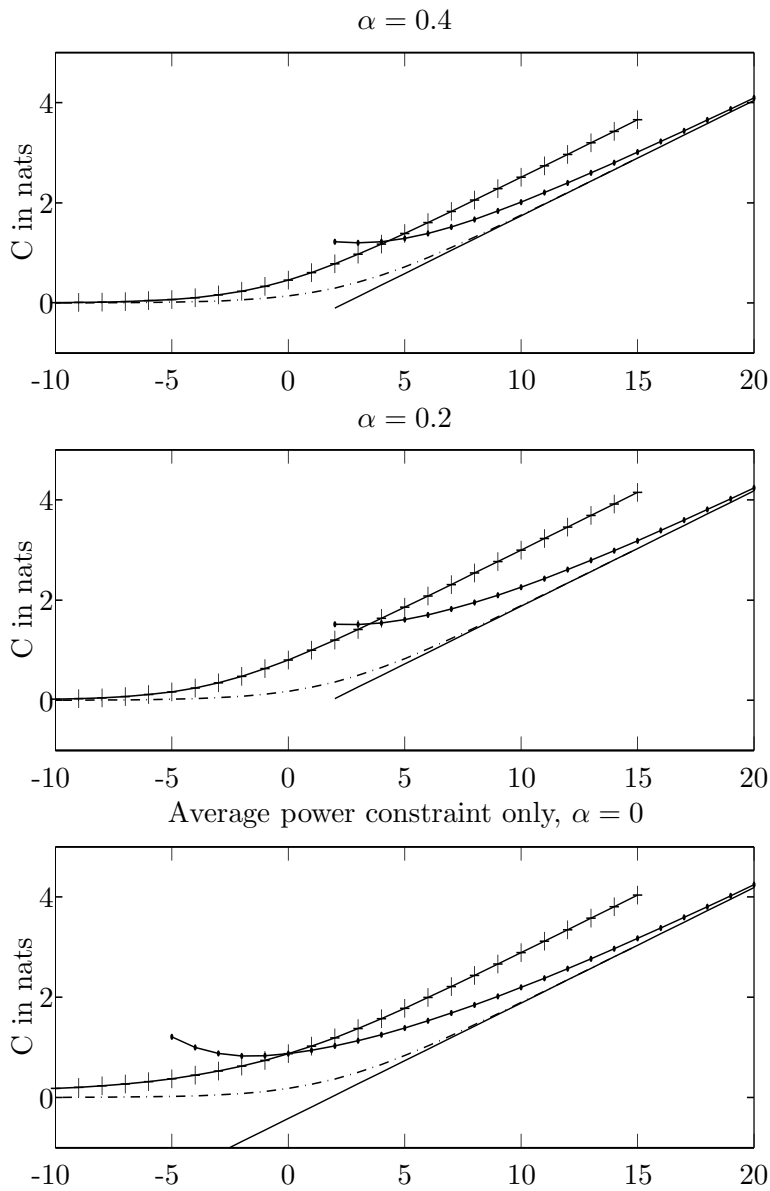

Figure 6.5: Evolution of the bounds for average and peak power constrained channels in function of the average power with decreasing α.

6.2 Comparison to Previous Results

The results for the channel capacity found in the previous sections are compared to some bounds in the literature. So far to our knowledge only two papers [1] and [2] that analyze the capacity of the free-space optical intensity channel have been published. The comparison to our bounds are actually not always straight forward, as in [1] an upper bound for the capacity of a band-limited channel has been computed. To solve this problem we will introduce a bandwidth constraint in our bound as well by applying the Nyquist-Shannon Theorem (see [4]).

6.2.1 Capacity Resulting from [1]

In [1] Roy You and Joseph M. Kahn present an upper bound on the channel capacity of the free-space optical intensity channel using pulse amplitude modulation (PAM) or quadrature amplitude modulation (QAM) (see [12]) schemes.

The channel they analyze is less general than the channel we have been looking at. They do not take into account the peak power constraint, but only the non-negativity of the input and the average optical power constraint. Additionally, they consider a bandwidth constraint on the channel. As they consider only PAM or QAM schemes actually not an upper bound for the channel capacity is found, but only an upper bound for the maximum mutual information between the input and the output of the channel that can be achieved when using PAM or QAM. These maximum mutual informations are of course smaller than the true capacity of the channel. We will see that nevertheless at high average optical power the upper bound for the maximum mutual information using a QAM scheme found by You and Kahn is larger than the true capacity.

The upper bounds presented in [1] for the maximum mutual information for a PAM and a QAM scheme, respectively, are:

$$C_{PAM} = W \log\left(\frac{P_{\text{avg}}}{\sigma}\right) + W \log\left(\frac{2d}{\sqrt{\pi e}}\right), \tag{6.6}$$

$$C_{QAM} = 2W \log\left(\frac{P_{\text{avg}}}{\sigma}\right) + W \log\left(\frac{1}{2}\right), \tag{6.7}$$

where W denotes the available bandwidth and the parameter d represents

the limit of the following expression:

$$d = \lim_{k\to\infty} \sqrt{k} \prod_{i=1}^{k} \left(\frac{2i}{2i+1} \right)^{k-i} \approx 1.46. \tag{6.8}$$

For PAM the ratio $\frac{P_{\text{avg}}}{\sigma}$ has to be larger than $\frac{\sqrt{\pi e}}{2d}$ and for QAM larger than $\sqrt{2}$, as otherwise the upper bound for the capacity becomes 0 and no reliable communication is possible using these modulation schemes.

We want to compare these bounds to the expression for the asymptotic capacity at high average power (4.10). In contrast to the bounds (6.7) and (6.6) the expression for the asymptotic capacity does not consider a bandwidth constraint. Therefore, we have to introduce such a constraint also in (4.10), in order to compare it to the bounds for PAM and QAM, respectively. We use the Nyquist-Shannon Theorem which states that with a signal of bandwidth W we have at most $2W$ degrees of freedom per second. So, if the bandwidth W is available we can use our channel $2W$ times per second and expression (4.10) turns into:

$$C = 2W \log \left(\frac{P_{\text{avg}}}{\sigma} \right) + W \log \left(\frac{e}{2\pi} \right) + o(1) \tag{6.9}$$

Comparing the expression for the capacity to the upper bound for PAM (6.6) we see that (6.9) grows much faster in P_{avg}. This is not really surprising as the expression for the capacity found in this report is actually a maximum of the mutual information over all the modulation schemes respecting the bandwidth W, whereas the bounds (6.6) and (6.7) hold only for a restricted set of signals, namely those modulated with PAM and QAM, respectively. It is more striking to notice that for high average power P_{avg} the upper bound for the maximal mutual information that can be obtained using a QAM scheme (6.7) is approximately by $0.1447 \cdot W$ nats per second higher than capacity (6.9). Figure 6.6 shows the bounds (6.6) and (6.7) and the capacity for the average power constrained channel (6.9) at high average power.

6.2.2 Capacity Resulting from [2]

In [2] Steve Hranilovic and Frank R. Kschischang present an upper and a lower bound for the channel capacity of the free-space optical intensity channel using general M-dimensional signals. The two bounds become tight for

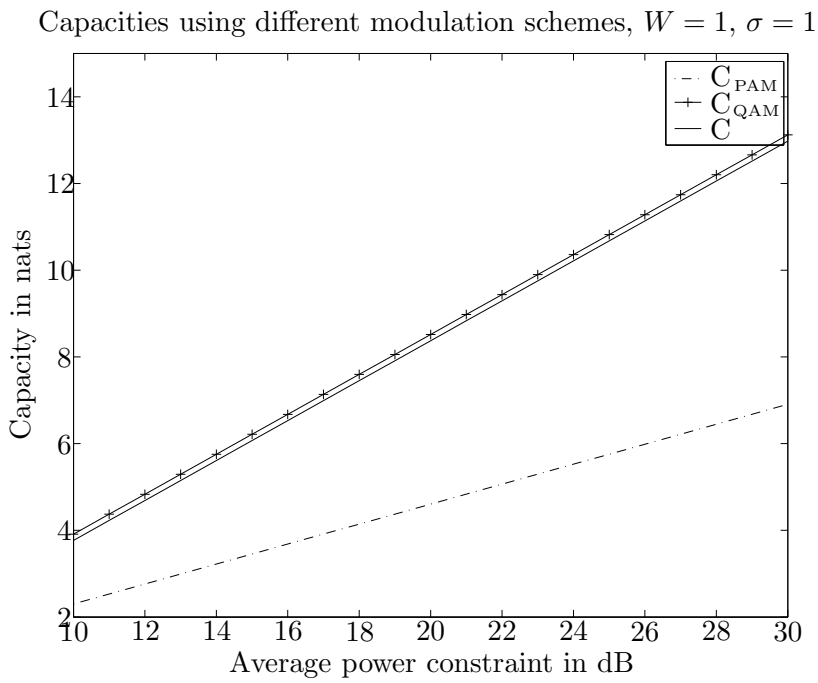

Figure 6.6: Capacity of the average power constrained channel compared to the results in [1].

high power. This paper is more general than the previous in the sense that the capacity is computed for all possible modulation schemes. But still the model is restricted as the peak power constraint is not considered.

Before presenting the result of this paper we have to introduce some notation. In [2] the transmitted signal $x(t)$ is represented as a linear combination of some orthonormal basis $\{\Phi_i(t)\}$ with dimension M:

$$x(t) = \sum_{i=1}^{M} x_i \Phi_i(t), \tag{6.10}$$

where the basis $\{\Phi_i(t)\}$ is chosen in such a way that all the optical power is contained in Φ_1. Further, the set of the M-dimensional vectors of the coefficients $\{x_i\}$ such that the resulting signal (6.10) is nonnegative for all times t and has average power k is denoted by Υ_k.

Additionally, in order to compare the expression for the capacity found in [2] with (4.10) at high average power, the constraint on the dimension of the basis has to be relaxed. The result in the paper of Hranilovic and

Kschischang is then:

$$C = \lim_{M\to\infty} C_{HK} = \lim_{M\to\infty} \log\left(\frac{P_{\text{avg}}}{\sigma}\sqrt{\frac{e}{2\pi}}\frac{V(\Upsilon_1)^{\frac{1}{M}}}{e}\right), \tag{6.11}$$

where $V(\Upsilon_1)$ denotes the volume of the set of coefficient-vectors which determine a nonnegative signal with power 1. Unfortunately, this expression could not be generalized in order to compute the capacity. The problem is that it was not possible to determine $V(\Upsilon_1)$ for the modulation scheme which achieves capacity. We can use now (4.10) in comparison to (6.11) to compute the asymptotic value for $V(\Upsilon_1)$ in the case where capacity is achieved:

$$\lim_{M\to\infty} V(\Upsilon_1)^{\frac{1}{M}} = e, \tag{6.12}$$

which means that $V(\Upsilon_1)$ grows exponentially in M.

Chapter 7

Conclusions and Outlook

The capacity of the free-space optical intensity channel, subject to an average and a peak power constraint, was investigated in this report, i.e. upper and lower bounds were computed. For reasons of simplicity at the beginning two simpler channel models were used. So, first some bounds were derived for the capacity of a channel which does not respect the peak power constraint. Then, the average power constraint was relaxed and the peak power constraint only was taken into account and last both power constraints were considered. The upper bounds for the capacity were found by using a dual expression for the capacity, whereas to obtain good lower bounds the common formula for the capacity together with the entropy power inequality was applied.

For all the three different channel models upper and lower bounds were found which become tight at high power. So, the asymptotic capacity at high power of the free-space optical intensity channel could be computed analytically exact. By comparing the asymptotic expressions for the capacity of the three different models it can be shown, that the error on the capacity arising from relaxing one of the power constraints in the complete channel model is smaller than 0.15 nats. This legitimates the choice of examining also the two simplified models. The advantage of these simplified models is that the expression for the asymptotic capacity has a closed form, in contrast to the capacity of the channel model respecting both power constraints.

At low power for the channel respecting a peak power constraint (with or without additional average power constraint) upper and lower bounds could be found which tend both to 0. In contrast to this, for the capacity of the

channel subject to only an average power constraint the best upper bound found tends to a small positive value (ca. 0.0811 nats).

One of the main objectives of this report was to examine if good upper bounds for the capacity of the free-space optical intensity channel could be found by applying the dual expression for the capacity (2.17). Indeed this was possible, even though for the cases where the peak power constraint is taken into account the same asymptotic results at high power can also be achieved by using the common expression for the capacity (2.11). However, these computations require a mathematically more difficult proof.

Comparing the results in this report with previous ones it can be seen that many new results were obtained. The most important result is the expression for the asymptotic channel capacity at high power for a general modulation scheme. Furthermore, some good upper and lower bounds at low power were obtained. To our knowledge so far no bounds in this power region were known.

However, the upper bound for the capacity in the low power region in the case of a channel with only average power constraint needs further improvement. The main problem is that we do not know how to upper bound the second moment of the capacity achieving input density. Therefore, it is not known whether with the dual formula an upper bound can be found which tends to 0 if the average power constraint tends to 0.

Furthermore, it is not clear whether the lower bounds represent the asymptotic behaviour of the channel capacity exactly. Probably, also for the lower bounds improvements are possible, e.g. by not applying the entropy power inequality but a better approximation.

Appendix A

Proofs of Chapter 4

A.1 Proof of (4.6)

Equation (4.6) presents an upper bound for the channel capacity with average power constraint $\mathrm{E}[X] \leq P_{\text{avg}}$. It is based on (2.17) and the choice of the output density of the Fisher-Tippett distribution:

$$R(y) = \frac{e^{\frac{a-y}{b} - e^{\frac{a-y}{b}}}}{b}. \tag{A.1}$$

Using this output density leads us to the following derivation:

$$\begin{aligned}
&\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-X)^2}{2\sigma^2}} \log \frac{e^{\frac{a-y}{b} - e^{\frac{a-y}{b}}}}{b} \,\mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-X)^2}{2\sigma^2}} \left(-\frac{y}{b} + \frac{a}{b} - \log b - e^{\frac{a-y}{b}}\right) \mathrm{d}y\right] \\
&= -\frac{a}{b} + \log b + \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-X)^2}{2\sigma^2}} \frac{y}{b} \mathrm{d}y\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-X)^2}{2\sigma^2}} e^{\frac{a}{b}} e^{-\frac{y}{b}} \mathrm{d}y\right] \\
&= -\frac{a}{b} + \log b + \mathrm{E}_{Q^*}\left[\frac{X}{b}\right] + e^{\frac{a}{b}} \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-X)^2}{2\sigma^2}} e^{-\frac{y}{b}} \mathrm{d}y\right] \\
&= -\frac{a}{b} + \log b + \frac{P_{\text{avg}}}{b} + e^{\frac{a}{b}} \mathrm{E}_{Q^*}\left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\left(y - \left(X - \frac{\sigma^2}{b}\right)\right)^2}{2\sigma^2}} \mathrm{d}y \; e^{-\frac{x}{b}} e^{\frac{\sigma^2}{2b^2}}\right] \\
&= -\frac{a}{b} + \log b + \frac{P_{\text{avg}}}{b} + e^{\frac{a}{b} + \frac{\sigma^2}{2b^2}} E_{Q^*}\left[e^{\frac{-X}{b}}\right] \\
&\leq -\frac{a}{b} + \log b + \frac{P_{\text{avg}}}{b} + e^{\frac{a}{b} + \frac{\sigma^2}{2b^2}}.
\end{aligned} \tag{A.2}$$

Therefore, we obtain an upper bound for the capacity for any value of the parameters a and b:

$$C \leq C_{\mathrm{U}} = \frac{-a}{b} + \log b + \frac{P_{\mathrm{avg}}}{b} + e^{\frac{a}{b}+\frac{\sigma^2}{2b^2}} - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \tag{A.3}$$

Now we have to choose the values for a and b. Suboptimally we will choose:

$$a = \frac{-\sigma^2}{2b} - k, \tag{A.4}$$

$$b = \frac{P_{\mathrm{avg}} + k + \sqrt{(P_{\mathrm{avg}}+k)^2 + 4\sigma^2}}{2}. \tag{A.5}$$

This yields the upper bound:

$$\begin{aligned} C_{\mathrm{U}} = & \frac{2(P_{\mathrm{avg}}+k)}{P_{\mathrm{avg}} + k + \sqrt{(P_{\mathrm{avg}}+k)^2 + 4\sigma^2}} \\ & + \frac{2\sigma^2}{\left(P_{\mathrm{avg}} + k + \sqrt{(P_{\mathrm{avg}}+k)^2 + 4\sigma^2}\right)^2} \\ & + \log\left(\frac{\mathrm{P}_{\mathrm{avg}} + k + \sqrt{(P_{\mathrm{avg}}+k)^2 + 4\sigma^2}}{2}\right) \\ & + e^{-k} - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{A.6}$$

To get a close upper bound for high average power we choose k growing sublinearly with P_{avg}:

$$k = \log P_{\mathrm{avg}}. \tag{A.7}$$

Then we obtain the following upper bound:

$$\begin{aligned} C_{\mathrm{U}} = & \frac{2(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})}{P_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}} \\ & + \log\left(\frac{\mathrm{P}_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}}{2}\right) \\ & + \frac{2\sigma^2}{\left(P_{\mathrm{avg}} + \log P_{\mathrm{avg}} + \sqrt{(P_{\mathrm{avg}} + \log P_{\mathrm{avg}})^2 + 4\sigma^2}\right)^2} + \frac{1}{P_{\mathrm{avg}}} \\ & - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{A.8}$$

A.2 Proof of (4.8)

Equation (4.8) presents an upper bound for the channel capacity with average power constraint. Again it is based on (2.17) and the choice of the

density of a double side exponential distribution:

$$R(y) = \begin{cases} \frac{p}{\beta} e^{-\frac{y-m}{\beta}}, & y \geq m, \\ \frac{1-p}{\alpha} e^{\frac{y-m}{\alpha}}, & y < m, \end{cases} \tag{A.9}$$

with parameters $m \leq 0$, $\alpha, \beta \geq 0$, $0 \leq p \leq 1$. We compute the first term of (2.17):

$$\begin{aligned} & \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X) \log R(y)\, \mathrm{d}y\right] \\ &= \underbrace{\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{m} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\frac{1-p}{\alpha} e^{\frac{y-m}{\alpha}}\right) \mathrm{d}y\right]}_{\mathrm{I}} \\ &\quad + \underbrace{\mathrm{E}_{Q^*}\left[-\int_{m}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\frac{p}{\beta} e^{-\frac{y-m}{\beta}}\right) \mathrm{d}y\right]}_{\mathrm{II}}. \end{aligned} \tag{A.10}$$

For expression I we can derive:

$$\begin{aligned} \mathrm{I} &= \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{m} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \left(\log\frac{1-p}{\alpha} - \frac{m}{\alpha}\right) \mathrm{d}y\right] \\ &\quad + \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{m} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \frac{y}{\alpha}\, \mathrm{d}y\right] \\ &= \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{m} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \left(\log\frac{1-p}{\alpha} - \frac{m}{\alpha}\right) \mathrm{d}y\right] \\ &\quad + \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{m-X} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}} \frac{z+X}{\alpha}\, \mathrm{d}z\right] \\ &\leq \mathrm{E}_{Q^*}\left[\left(1 - \mathrm{Q}\left(\frac{m-X}{\sigma}\right)\right) \log\frac{\alpha}{1-p}\right] \\ &\quad + \mathrm{E}_{Q^*}\left[\left(1 - \mathrm{Q}\left(\frac{m-X}{\sigma}\right)\right) \frac{m-X}{\alpha}\right] + \mathrm{E}_{Q^*}\left[\frac{\sigma}{\sqrt{2\pi}\alpha} e^{-\frac{(m-X)^2}{2\sigma^2}}\right] \\ &\leq \mathrm{E}_{Q^*}\left[\left(1 - \mathrm{Q}\left(\frac{m-X}{\sigma}\right)\right) \log\frac{\alpha}{1-p}\right] + \frac{\sigma}{\sqrt{2\pi}\alpha} e^{-\frac{m^2}{2\sigma^2}}, \end{aligned} \tag{A.11}$$

where the two inequalities follow from the fact that $\frac{m}{\alpha}$ and $\frac{m-X}{\alpha}$, respectively, are negative. We will choose α and p such that $\log\frac{\alpha}{1-p} \leq 0$. The new upper bound for I is then:

$$\mathrm{I} \leq \frac{\sigma}{\sqrt{2\pi}\alpha} e^{-\frac{m^2}{2\sigma^2}}. \tag{A.12}$$

Next, we want to find an upper bound for the second term:

$$\begin{aligned}
\mathrm{II} &= \mathrm{E}_{Q^*}\left[\int_m^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \left(\log\beta - \log p - \frac{m}{\beta}\right) \mathrm{d}y\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\int_m^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \frac{y}{\beta} \mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[\mathrm{Q}\left(\frac{m-X}{\sigma}\right)\left(\log\beta - \log p - \frac{m}{\beta}\right)\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\int_{m-X}^\infty \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}} \frac{z+X}{\beta} \mathrm{d}z\right] \\
&= \mathrm{E}_{Q^*}\left[\mathrm{Q}\left(\frac{m-X}{\sigma}\right)\left(\log\beta - \log p - \frac{m-X}{\beta}\right)\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\frac{\sigma}{\sqrt{2\pi}\beta} e^{-\frac{(m-X)^2}{2\sigma^2}}\right] \\
&\leq \mathrm{E}_{Q^*}\left[\mathrm{Q}\left(\frac{m-X}{\sigma}\right)\left(\log\beta - \log p - \frac{m-X}{\beta}\right)\right] + \frac{\sigma}{\sqrt{2\pi}\beta} e^{-\frac{m^2}{2\sigma^2}}
\end{aligned} \tag{A.13}$$

Here we will choose $\beta \geq 1$. With this we get a new upper bound for the second term:

$$\mathrm{II} \leq \log\beta - \log p + \frac{P_{\mathrm{avg}} - m}{\beta} + \frac{\sigma}{\sqrt{2\pi}\beta} e^{-\frac{m^2}{2\sigma^2}}. \tag{A.14}$$

The upper bound for the capacity is then:

$$\begin{aligned}
C_{\mathrm{U}} &= \log\beta - \log p + \frac{P_{\mathrm{avg}} - m}{\beta} + \frac{\sigma}{\sqrt{2\pi}\beta} e^{-\frac{m^2}{2\sigma^2}} + \frac{\sigma}{\sqrt{2\pi}\alpha} e^{-\frac{m^2}{2\sigma^2}} \\
&\quad - \frac{1}{2}\log\left(2\pi e\sigma^2\right),
\end{aligned} \tag{A.15}$$

where the parameters have to fulfill the following conditions:

$$m \leq 0, \tag{A.16}$$

$$\beta \geq 1, \tag{A.17}$$

$$0 \leq \alpha \leq 1 - p, \tag{A.18}$$

$$0 \leq p \leq 1. \tag{A.19}$$

Instead of minimizing C_{U} over all the parameters we choose suboptimally:

$$m = -k\sigma, \tag{A.20}$$

$$\alpha = 1 - p = \frac{1}{k} \tag{A.21}$$

$$\text{and } k \geq 1. \tag{A.22}$$

Then, we minimize C_{U} over β by taking the derivative and setting it equal to 0. With the second derivative it can be verified, that indeed the minimal value is received. The obtained optimal value for β is:

$$\beta = \frac{\sigma}{\sqrt{2\pi}} e^{-\frac{k^2}{2}} + P_{\mathrm{avg}} + k\sigma. \tag{A.23}$$

By choosing $k \geq \frac{1}{\sigma}$ the condition (A.17) is fulfilled. And we get the following upper bound for the capacity:

$$\begin{aligned} C \leq C_{\mathrm{U}} &= \log\left(\frac{\sigma}{\sqrt{2\pi}} e^{-\frac{k^2}{2}} + P_{\mathrm{avg}} + k\sigma\right) - \log\left(1 - \frac{1}{k}\right) + 1 + \frac{\sigma k}{\sqrt{2\pi}} e^{-\frac{k^2}{2}} \\ &\quad - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \end{aligned} \tag{A.24}$$

with the condition:

$$k \geq \max\left\{1, \frac{1}{\sigma}\right\}. \tag{A.25}$$

To get a small upper bound for high average power we want k tending to infinity, sublinearly with P_{avg}. Therefore, we choose $k = \log P_{\mathrm{avg}}$:

$$\begin{aligned} C_{\mathrm{U}} &= \log\left(\frac{\sigma}{\sqrt{2\pi}} e^{-\frac{(\log P_{\mathrm{avg}})^2}{2}} + P_{\mathrm{avg}} + \sigma \log P_{\mathrm{avg}}\right) - \log\left(1 - \frac{1}{\log P_{\mathrm{avg}}}\right) \\ &\quad + 1 + \frac{\sigma \log P_{\mathrm{avg}}}{\sqrt{2\pi}} e^{-\frac{(\log P_{\mathrm{avg}})^2}{2}} - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \end{aligned} \tag{A.26}$$

where

$$P_{\mathrm{avg}} \geq \max\left\{e, e^{\frac{1}{\sigma}}\right\}. \tag{A.27}$$

A.3 Proof of (4.18)

Equation (4.18) presents an upper bound for the channel capacity with peak power constraint A based on the dual expression (2.17) and a choice of the output density

$$R(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y < -\delta, \\ \frac{P^*}{A+2\delta}, & -\delta \leq y \leq A + \delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y > A + \delta, \end{cases}$$

where:

$$\delta \geq 0, \tag{A.28}$$

and P^* is the probability that $y \in [-\delta, A+\delta]$:

$$\begin{aligned} P^* &= 1 - \int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \mathrm{d}y - \int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \mathrm{d}y \\ &= \mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right). \end{aligned} \tag{A.29}$$

This output density leads to the following expression:

$$\begin{aligned} &\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X) \log R(y) \mathrm{d}y\right] \\ &= \underbrace{\mathrm{E}_{Q^*}\left[\int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right) \mathrm{d}y\right]}_{\mathrm{I}} \\ &\quad + \underbrace{\mathrm{E}_{Q^*}\left[-\int_{-\delta}^{A+\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log \frac{P^*}{A+2\delta} \mathrm{d}y\right]}_{\mathrm{II}} \\ &\quad + \underbrace{\mathrm{E}_{Q^*}\left[\int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{(y-A)^2}{2\sigma^2}}\right) \mathrm{d}y\right]}_{\mathrm{III}}. \end{aligned} \tag{A.30}$$

First, we upper bound the expressions I and III. Therefore we will need the following lemma:

Lemma A.1 *For an AWGN channel with input between* 0 *and* A *the transition probability* $W(y|x)$ *for* $y \in [-\infty, -\delta]$ *can be upper bounded by:*

$$W(y|x) \leq \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, \quad y \leq -\delta \tag{A.31}$$

and for $y \in [A+\delta, \infty]$ *by:*

$$W(y|x) \leq \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, \quad y \geq A+\delta, \tag{A.32}$$

where $\delta \geq 0$.

Proof: The transition probability

$$W(y|x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}} \tag{A.33}$$

is a function of the distance between y and x and decreases when this distance increases. Therefore $W(y|x)$ is maximized for any y by choosing the value $x \in [0, A]$ which is nearest to y. For $y \geq A$ this is $x_{\text{nearest}} = A$ and for

$y \leq 0$ this is $x_{\text{nearest}} = 0$. $\square$

Applying Lemma A.1 we obtain some bounds for I and III:

$$\begin{aligned} \mathrm{I} &\leq \mathrm{E}_{Q^*}\left[\int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right) \mathrm{d}y\right] \\ &= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2}\right) + \frac{\delta}{2\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}}, \end{aligned} \tag{A.34}$$

$$\begin{aligned} \mathrm{III} &\leq \mathrm{E}_{Q^*}\left[\int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{(y-A)^2}{2\sigma^2}}\right) \mathrm{d}y\right] \\ &= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2}\right) + \frac{\delta}{2\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}}. \end{aligned} \tag{A.35}$$

Then, we compute II:

$$\begin{aligned} \mathrm{II} &= \mathrm{E}_{Q^*}\left[\int_{-\delta}^{A+\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\frac{A+2\delta}{P^*} \mathrm{d}y\right] \\ &= \mathrm{E}_{Q^*}\left[\left(1 - \mathcal{Q}\left(\frac{\delta+X}{\sigma}\right) - \mathcal{Q}\left(\frac{A+\delta-X}{\sigma}\right)\right) \log\frac{A+2\delta}{P^*}\right] \\ &\leq \log(A+2\delta) - \log P^*. \end{aligned} \tag{A.36}$$

Combining I, II, III and (A.29) we get an upper bound for the capacity of the channel with peak power constraint A:

$$\begin{aligned} C \leq\ & \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\delta}{\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}} \\ &+ \log(A+2\delta) - \log\left(\mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\right) \\ &- \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{A.37}$$

To get a good upper bound at high peak power δ has to grow sublinearly with the peak power A. Therefore, we set:

$$\delta = \log A, \tag{A.38}$$

and we obtain a new upper bound for the capacity under the condition that (A.28) is still fulfilled:

$$\begin{aligned} C \leq C_{\mathrm{U}} =\ & \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma} e^{-\frac{(\log A)^2}{2\sigma^2}} \\ &+ \log(A + 2\log A) - \log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right) - \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right) \\ &- \frac{1}{2}\log\left(2\pi e\sigma^2\right). \end{aligned} \tag{A.39}$$

A.4 Proof of (4.30)

Equation (4.30) presents an upper bound for the channel capacity with average and peak power constraints P_{avg} and A based on the dual expression (2.17) and the choice for the output density:

$$R(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y \leq -\delta, \\ \frac{P^*\lambda}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}} e^{-\lambda y}, & -\delta \leq y \leq A+\delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y \geq A+\delta, \end{cases}$$

where:

$$\delta \geq 0, \tag{A.40}$$

and P^* is the probability that $y \in [-\delta, A+\delta]$:

$$\begin{aligned} P^* &= 1 - \int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \mathrm{d}y - \int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \mathrm{d}y \\ &= \mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right). \end{aligned} \tag{A.41}$$

This choice of $R(y)$ yields:

$$\begin{aligned} &\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X) \log R(y) \, \mathrm{d}y\right] \\ &= \underbrace{\mathrm{E}_{Q^*}\left[\int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right) \mathrm{d}y\right]}_{\text{I}} \\ &\quad + \underbrace{\mathrm{E}_{Q^*}\left[-\int_{-\delta}^{A+\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\frac{P^*\lambda}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}} e^{-\lambda y}\right) \mathrm{d}y\right]}_{\text{II}} \\ &\quad + \underbrace{\mathrm{E}_{Q^*}\left[\int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{(y-A)^2}{2\sigma^2}}\right) \mathrm{d}y\right]}_{\text{III}}, \end{aligned} \tag{A.42}$$

where the expressions I and III already have been computed in (A.34) and (A.35). Therefore, what remains to compute is the expression II:

$$\begin{aligned}
\mathrm{II} &= \mathrm{E}_{Q^*}\left[\int_{-\delta}^{A+\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{P^*\lambda} e^{\lambda y}\right) \mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[\left(\mathcal{Q}\left(\frac{-\delta - X}{\sigma}\right) - \mathcal{Q}\left(\frac{A+\delta-X}{\sigma}\right)\right) \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{P^*\lambda}\right)\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\int_{-\delta}^{A+\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \lambda y \mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[\underbrace{\left(1 - \mathcal{Q}\left(\frac{\delta+X}{\sigma}\right) - \mathcal{Q}\left(\frac{A+\delta-X}{\sigma}\right)\right)}_{\leq 1} \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{P^*\lambda}\right)\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\underbrace{\left(1 - \mathcal{Q}\left(\frac{\delta+X}{\sigma}\right) - \mathcal{Q}\left(\frac{A+\delta-X}{\sigma}\right)\right)}_{\leq 1} \lambda X\right] \\
&\quad + \mathrm{E}_{Q^*}\left[\frac{\lambda\sigma}{\sqrt{2\pi}}\left(e^{-\frac{(\delta+X)^2}{2\sigma^2}} - e^{-\frac{(A+\delta-X)^2}{2\sigma^2}}\right)\right] \\
&\leq \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{P^*\lambda}\right) + \lambda P_{\mathrm{avg}} + \frac{\lambda\sigma}{\sqrt{2\pi}}\left(e^{-\frac{\delta^2}{2\sigma^2}} - e^{-\frac{(A+\delta)^2}{2\sigma^2}}\right). \qquad \text{(A.43)}
\end{aligned}$$

Combining I, II, III and (A.41) we get an upper bound for the capacity:

$$\begin{aligned}
C \leq C_{\mathrm{U}} &= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\delta}{\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}} \\
&\quad + \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{\lambda}\right) - \log\left(\mathcal{Q}\left(-\frac{\delta}{\sigma}\right) - \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\right) \\
&\quad + \lambda P_{\mathrm{avg}} + \frac{\lambda\sigma}{\sqrt{2\pi}}\left(e^{-\frac{\delta^2}{2\sigma^2}} - e^{-\frac{(A+\delta)^2}{2\sigma^2}}\right) \\
&\quad - \frac{1}{2}\log\left(2\pi e\sigma^2\right). \qquad \text{(A.44)}
\end{aligned}$$

Setting the parameters λ and δ to:

$$\delta = \log A, \qquad \text{(A.45)}$$

$$\lambda = \frac{k}{A}, \qquad \text{(A.46)}$$

we get a new upper bound for the capacity with average power constraint P_{avg} and peak power constraint A under the condition that (A.40) is still

fulfilled:

$$\begin{aligned} C \leq C_{\mathrm{U}} \quad = \quad & \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}} \\ & +\log\left(\frac{e^{\frac{k\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{k}\right)+\log(A) \\ & -\log\left(\mathcal{Q}\left(-\frac{\log A}{\sigma}\right)-\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\right)+k\frac{P_{\mathrm{avg}}}{A} \\ & +\frac{k\sigma}{\sqrt{2\pi}A}\left(e^{-\frac{(\log A)^2}{2\sigma^2}}-e^{-\frac{(A+\log A)^2}{2\sigma^2}}\right)-\frac{1}{2}\log\left(2\pi e\sigma^2\right), \end{aligned} \tag{A.47}$$

where k is a parameter and can have any non-negative value.

Appendix B

Alternative Upper Bounds for Peak Power Constrained Channels

B.1 Alternative Upper Bound for the Capacity with only Peak Power Constraint at High Peak Power

In this section we are going to present you an alternative way of deriving an upper bound for the capacity of the channel with only peak power constraint. We will see that for high peak power the result will be the same as for the result given in Section 4.3. However, with this alternative proof we hope to give a little bit more of intuition about how we found the channel capacity with peak power constraint.

To upper bound the capacity we use (2.11), but perform the maximization over all the possible output densities, instead of over all the input densities:

$$C = \max_{f(y)\in\mathcal{P}(\mathcal{Y})_A} \Big\{ h(Y) - h(Y|X) \Big\}, \tag{B.1}$$

where $P(\mathcal{Y})_A$ denotes the set of all output probability densities induced by the channel from some input density with peak constraint A.

Lemma B.1 *For an AWGN channel with input between* 0 *and* A *the output*

density $f(y)$ can be upper bounded in the interval $[-\infty, -\delta]$ by:

$$f(y) \leq \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, \quad y \leq -\delta, \tag{B.2}$$

and in the interval $[A+\delta, \infty]$ by:

$$f(y) \leq \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, \quad y \geq A+\delta, \tag{B.3}$$

where $\delta \geq 0$.

Proof: Lemma A.1 yields an upper bound for the output density:

$$\begin{aligned} f(y) &= \int_{x\in\mathcal{X}} Q(x)W(y|x)\mathrm{d}x = \int_{x\in\mathcal{X}} Q(x)\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}}\mathrm{d}x \\ &\leq \int_{x\in\mathcal{X}} Q(x)\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}\mathrm{d}x = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}. \end{aligned} \tag{B.4}$$

Similarly, for $y \in [A+\delta, \infty]$ we obtain:

$$\begin{aligned} f(y) &= \int_{x\in\mathcal{X}} Q(x)W(y|X)\mathrm{d}x = \int_{x\in\mathcal{X}} Q(x)\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-x)^2}{2\sigma^2}}\mathrm{d}x \\ &\leq \int_{x\in\mathcal{X}} Q(x)\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}\mathrm{d}x = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}. \end{aligned} \tag{B.5}$$

□

Thus, by applying Lemma B.1 we can upper bound the capacity achieving output density:

$$f^*(y) \leq f_{\mathrm{U}}(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y < -\delta, \\ f^*(y), & -\delta \leq y \leq A+\delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y > A+\delta. \end{cases} \tag{B.6}$$

Here $f^*(y)$ denotes the true capacity achieving output density, which we do not know so far.

Lemma B.2 *For two nonnegative functions $f(y)$ and $f'(y)$, where $f'(y) \geq f(y)$ and $f'(y) \leq e^{-1}$, for $y \in [a, b]$, holds:*

$$-\int_a^b f(y)\log f(y)\mathrm{d}y \leq -\int_a^b f'(y)\log f'(y)\mathrm{d}y. \tag{B.7}$$

Proof: The expression $-x \log x$ is monotonically increasing in the interval $[0, e^{-1}]$. □

Under the condition:

$$\delta \geq \sqrt{2\sigma^2 \log\left(\frac{e}{\sqrt{2\pi}\sigma}\right)} \tag{B.8}$$

we have

$$f_{\mathrm{U}}(y) \leq e^{-1}, \quad y \in [-\infty, -\delta] \quad \text{and} \quad y \in [A+\delta, \infty] \tag{B.9}$$

and we can apply Lemma B.2 to the differential output entropy in (B.1):

$$\begin{aligned}
\max_{f(y)\in\mathcal{P}(\mathcal{Y})_A} h(Y) &= h_{f^*}(Y) \leq h_{f_{\mathrm{U}}}(Y) = -\int_{-\infty}^{\infty} f_{\mathrm{U}}(y) \log f_{\mathrm{U}}(y)\,\mathrm{d}y \\
&= \underbrace{\int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right) \mathrm{d}y}_{\mathrm{I}} \\
&\quad + \underbrace{\int_{-\delta}^{A+\delta} -f^*(y) \log f^*(y)\,\mathrm{d}y}_{\mathrm{II}} \\
&\quad + \underbrace{\int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right) \mathrm{d}y}_{\mathrm{III}}.
\end{aligned} \tag{B.10}$$

In (A.34) and (A.35) we have already found:

$$\mathrm{I} = \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2}\right) + \frac{\delta}{2\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}}, \tag{B.11}$$

$$\mathrm{III} = \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2}\right) + \frac{\delta}{2\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}}. \tag{B.12}$$

To upper bound the expression II we substitute $f^*(y)$ by $P^* g^*(y)$:

$$\begin{aligned}
\mathrm{II} &= -\int_{-\delta}^{A+\delta} f^*(y) \log f^*(y)\,\mathrm{d}y \\
&= -P^* \int_{-\delta}^{A+\delta} g^*(y) \log g^*(y)\,\mathrm{d}y - P^* \log P^* \\
&= P^* h_{g^*}(Y) - P^* \log P^*,
\end{aligned} \tag{B.13}$$

where:

$$P^* = \int_{-\delta}^{A+\delta} f^*(y)\,\mathrm{d}y. \tag{B.14}$$

The advantage of this substitution is, that in the interval $[-\delta, A+\delta]$ $g^*(y)$ is a true probability density that integrates to 1. Therefore, we can upper bound (B.13) by maximizing the differential entropy over all probability densities in the interval $[-\delta, A+\delta]$:

$$\begin{aligned} \mathrm{II} &= P^* h_{g^*}(Y) - P^* \log P^* \leq \max_{g(y)\in\mathcal{P}(\mathcal{Y}_{[-\delta,A+\delta]})} P^* h_g(Y) - P^* \log P^* \\ &= P^* h_{g,\max}(Y) - P^* \log P^* \end{aligned} \tag{B.15}$$

This means that we have to find the density in the interval $[-\delta, A+\delta]$ which maximizes differential entropy. As we know already from Corollary 2.5 this is the density of the uniform distribution:

$$g_{\max}(y) = \frac{1}{A+2\delta}, \tag{B.16}$$

and the maximal differential entropy is:

$$h_{g,\max}(Y) = -\int_{-\delta}^{A+\delta} g_{\max}(y) \log g_{\max}(y) \mathrm{d}y = \log(A+2\delta). \tag{B.17}$$

This leads us to an upper bound for II:

$$\mathrm{II} \leq P^* \log(A+2\delta) - P^* \log P^*. \tag{B.18}$$

Now, we have an upper bound for the capacity of the channel with peak power constraint:

$$\begin{aligned} C &\leq \mathrm{I} + \mathrm{II} + \mathrm{III} - h(Y|X) \\ &\leq \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\delta}{\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}} \\ &\quad + P^* \log(A+2\delta) - P^* \log P^* - \frac{1}{2}\log\left(2\pi e \sigma^2\right) \end{aligned} \tag{B.19}$$

under the condition (B.8). With the choice:

$$\delta = \log A \tag{B.20}$$

we get a new upper bound for the capacity:

$$\begin{aligned} C \leq C_{\mathrm{U}} &= \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma} e^{-\frac{(\log A)^2}{2\sigma^2}} \\ &\quad + P^* \log(A + 2\log A) - P^* \log P^* \\ &\quad - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \end{aligned} \tag{B.21}$$

that due to condition (B.8) holds for $A \geq \exp\left(\sqrt{2\sigma^2 \log\left(\frac{e}{\sqrt{2\pi}\sigma}\right)}\right)$.

Unfortunately in (B.21) we cannot determine P^*, as still we do not know the capacity achieving output density $f^*(y)$. However, we know the asymptotic value for P^* when A (and also δ) tends to infinity:

$$\lim_{A\to\infty} P^* = \lim_{A\to\infty} \int_{-\log A}^{A+\log A} f^*(y) = 1. \tag{B.22}$$

Thus, we can find an asymptotic upper bound for the capacity.

Lemma B.3 *The capacity of the free-space optical intensity channel with peak power constraint only is upper bounded by:*

$$C_{\mathrm{U}} = \log A - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1), \tag{B.23}$$

where the o(1) term tends to 0 as the peak power tends to infinity.

Proof: With (B.21) and (B.22) the asymptotic upper bound (B.23) can be verified:

$$\begin{aligned}
&\lim_{A\to\infty}\left\{C_{\mathrm{U}} - \left(\log A - \frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\} \\
&= \lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}}\right. \\
&\quad + P^*\log(A + 2\log A) - P^*\log P^* - \frac{1}{2}\log\left(2\pi e\sigma^2\right) \\
&\quad \left. - \left(\log A - \frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\} \\
&= \lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right)\right. \\
&\quad \left. + \frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}} + \log\left(1 + 2\frac{\log A}{A}\right)\right\} \\
&= 0 + 0 + \log 1 = 0.
\end{aligned} \tag{B.24}$$

□

B.2 Alternative Upper Bound for the Capacity with Average and Peak Power Constraint at High Peak Power and fixed Average-to-Peak Power Ratio

In this section we present an alternative proof of the upper bound for the capacity of the channel with average and peak power constraint at high power. The computations are done in a similar way as in Section B.1. With this proof we hope to give a little bit more of intuition about how we found the channel capacity in Section 4.4.2.

To upper bound the capacity we use (B.1), meaning that we perform a maximization over the set of output probability densities induced by the channel from some input density with peak constraint A and average P_{avg}. Lemma B.1 leads to an upper bound for the capacity achieving output density:

$$f^*(y) \leq f_{\text{U}}(y) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & y < -\delta, \\ f^*(y), & -\delta \leq y \leq A + \delta, \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}, & y > A + \delta, \end{cases} \tag{B.25}$$

where $f^*(y)$ denotes the true capacity achieving output density, which we do not know so far.

Under the condition

$$\delta \geq \sqrt{2\sigma^2 \log\left(\frac{e}{\sqrt{2\pi}\sigma}\right)}, \tag{B.26}$$

we have:

$$f_{\text{U}}(y) \leq e^{-1}, \quad y \in [-\infty, -\delta] \quad \text{and} \quad y \in [A + \delta, \infty] \tag{B.27}$$

and we can apply Lemma B.2 to upper bound the first term of (B.1):

$$\begin{aligned}
\max_{f(y)\in P(\mathcal{Y})_A} h(Y) = h_{f^*}(Y) \leq h_{f_\mathrm{U}}(Y) &= -\int_{-\infty}^{\infty} f_\mathrm{U}(y)\log f_\mathrm{U}(y)\,\mathrm{d}y \\
= \underbrace{\int_{-\infty}^{-\delta} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{y^2}{2\sigma^2}}\right)\mathrm{d}y}_{\mathrm{I}} \\
\underbrace{-\int_{-\delta}^{A+\delta} f^*(y)\log f^*(y)\,\mathrm{d}y}_{\mathrm{II}} \\
+\underbrace{\int_{A+\delta}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}} \log\left(\sqrt{2\pi}\sigma e^{\frac{(y-A)^2}{2\sigma^2}}\right)\mathrm{d}y}_{\mathrm{III}}. \qquad \text{(B.28)}
\end{aligned}$$

In (A.34) and (A.35) we have already found:

$$\begin{aligned}
\mathrm{I} &= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right)+\frac{1}{2}\right)+\frac{\delta}{2\sqrt{2\pi}\sigma}e^{-\frac{\delta^2}{2\sigma^2}}, \qquad \text{(B.29)} \\
\mathrm{III} &= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(\log\left(\sqrt{2\pi}\sigma\right)+\frac{1}{2}\right)+\frac{\delta}{2\sqrt{2\pi}\sigma}e^{-\frac{\delta^2}{2\sigma^2}}. \qquad \text{(B.30)}
\end{aligned}$$

To upper bound the expression II we substitute $f^*(y)$ by $P^*g^*(y)$:

$$\begin{aligned}
\mathrm{II} &= -\int_{-\delta}^{A+\delta} f^*(y)\log f^*(y)\,\mathrm{d}y \\
&= -P^*\int_{-\delta}^{A+\delta} g^*(y)\log g^*(y)\,\mathrm{d}y - P^*\log P^* \\
&= P^* h_{g^*} - P^*\log P^*, \qquad \text{(B.31)}
\end{aligned}$$

where:

$$P^* = \int_{-\delta}^{A+\delta} f^*(y)\,\mathrm{d}y. \qquad \text{(B.32)}$$

Here $g^*(y)$ represents a true probability density in the interval $[-\delta, A+\delta]$ with expectation value:

$$\mathrm{E}_{g^*}[Y] = \int_{-\delta}^{A+\delta} g^*(y)y\,\mathrm{d}y = \frac{1}{P^*}\int_{-\delta}^{A+\delta} f^*(y)y\,\mathrm{d}y = \frac{\mathrm{E}_{f^*}[Y]}{P^*}. \qquad \text{(B.33)}$$

We want to upper bound $h_{g^*}(Y)$ by the maximal differential entropy of the densities in the interval $[-\delta, A+\delta]$ respecting the average constraint $\mathrm{E}_{g,\max}[Y] = \mathrm{E}_{g^*}[Y]$. From Theorem 2.2 we know that to achieve this we have to use the density of a truncated exponential distribution:

$$g_\max(y) = \frac{\lambda e^{-\lambda y}}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}, \quad y \in [-\delta, A+\delta] \qquad \text{(B.34)}$$

with expectation value:

$$\mathrm{E}_{g,\max}[Y] = \frac{1}{\lambda} - \frac{e^{-\lambda(A+\delta)}(A+\delta) + e^{\lambda\delta}\delta}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}. \tag{B.35}$$

This leads us to the following upper bound for II:

$$\begin{aligned}
\mathrm{II} &\leq P^* h_{g^*}(Y) - P^* \log P^* \leq \max_{g(y)\in\mathcal{P}(\mathcal{Y}_{[-\delta,A+\delta]})} P^* h_g(Y) - P^* \log P^* \\
&= P^* h_{g,\max}(Y) - P^* \log P^* \\
&= P^* \int_{-\delta}^{A+\delta} \frac{\lambda e^{-\lambda y}}{e^{\lambda\delta} - e^{-\lambda(A+\delta)}} \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{\lambda} e^{\lambda y}\right) \mathrm{d}y - P^* \log P^* \\
&= P^* \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{\lambda}\right) P^* + \lambda \mathrm{E}_{g,\max}[Y] - P^* \log P^*. \qquad \text{(B.36)}
\end{aligned}$$

Hence, we have an upper bound for the capacity of the channel with average and peak power constraints:

$$\begin{aligned}
C &\leq \mathrm{I} + \mathrm{II} + \mathrm{III} - h(Y|X) \\
&= \mathcal{Q}\left(\frac{\delta}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\delta}{\sqrt{2\pi}\sigma} e^{-\frac{\delta^2}{2\sigma^2}} \\
&\quad + P^* \log\left(\frac{e^{\lambda\delta} - e^{-\lambda(A+\delta)}}{\lambda}\right) + P^* \lambda \mathrm{E}_{g,\max}[Y] - P^* \log P^* \\
&\quad - \frac{1}{2}\log\left(2\pi e \sigma^2\right), \qquad \text{(B.37)}
\end{aligned}$$

where P^*, $\mathrm{E}_{g,\max}[Y]$ and λ depend on the average power constraint P_{avg} and additionally (B.26) must hold.

With the choice:

$$\delta = \log A \tag{B.38}$$

$$\lambda = \frac{k}{A} \tag{B.39}$$

we get a new upper bound for the capacity:

$$\begin{aligned}
C \leq C_{\mathrm{U}} &= \mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1 + 2\log\left(\sqrt{2\pi}\sigma\right)\right) + \frac{\log A}{\sqrt{2\pi}\sigma} e^{-\frac{(\log A)^2}{2\sigma^2}} \\
&\quad + P^* \log\left(\frac{e^{k\frac{\log A}{A}} - e^{-k(1+\frac{\log A}{A})}}{k}\right) + P^* \log A \\
&\quad + P^* k \frac{\mathrm{E}_{g,\max}[Y]}{A} - P^* \log P^* - \frac{1}{2}\log\left(2\pi e \sigma^2\right). \qquad \text{(B.40)}
\end{aligned}$$

We want to examine the upper bound for high peak power constraint A, where the ratio $\alpha = \frac{P_{\text{avg}}}{A}$ stays fixed. First, we investigate what happens to P^* when A and therefore also $\delta = \log A$ tend to infinity:

$$\lim_{A\to\infty} P^* = \lim_{A\to\infty} \int_{-\log A}^{A+\log A} f^*(y)\,\mathrm{d}y = 1. \tag{B.41}$$

Therefore, the distribution $g^*(y)$ tends to $f^*(y)$ with increasing A. Furthermore, $f_{\text{U}}(y)$ tends to $f^*(y)$ too, since the two intervals vanish where they are different. Consequently:

$$\begin{aligned}
\lim_{A\to\infty} \mathrm{E}_{g,\max}[Y] &= \lim_{A\to\infty} \mathrm{E}_{g^*}[Y] = \lim_{A\to\infty} \mathrm{E}_{f^*}[Y], && \text{(B.42)}\\
\lim_{A\to\infty} \mathrm{E}_{f_{\text{U}}}[Y] &= \lim_{A\to\infty} \mathrm{E}_{f^*}[Y], && \text{(B.43)}\\
\mathrm{E}_{f^*}[Y] &= P_{\text{avg}}, && \text{(B.44)}
\end{aligned}$$

where the last equality follows from Lemma 3.1. Thus, (B.42) and (B.44) tell us how to consider the average constraint in (B.40) for high peak power.

Lemma B.4 *The capacity of the free-space optical intensity channel with average and peak power constraints and fixed ratio $\alpha = \frac{P_{\text{avg}}}{A}$ is upper bounded by:*

$$C_{\text{U}} = \log A + \log\left(\frac{1-e^{-k}}{k}\right) + k\alpha - \frac{1}{2}\log\left(2\pi e\sigma^2\right) + o(1), \tag{B.45}$$

where the o(1) term tends to 0 as the peak power constraint tends to infinity. Additionally, the parameter k has to fulfill the following condition:

$$\alpha = \frac{1}{k} - \frac{e^{-k}}{1-e^{-k}}. \tag{B.46}$$

Proof: With (B.41), (B.42) and (B.44) it can be verified that the upper bound (B.40) for fixed ratio α and high peak power A asymptotically tends

to (B.45):

$$
\begin{aligned}
&\lim_{A\to\infty}\left\{C_U-\left(\log A+\log\left(\frac{1-e^{-k}}{k}\right)+k\alpha-\frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\}\\
&=\lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}}\right.\\
&\quad+P^*\log\left(\frac{e^{k\frac{\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{k}\right)+P^*\log A\\
&\quad+P^*k\frac{\mathrm{E}_{g,\max}[Y]}{A}-P^*\log P^*-\frac{1}{2}\log\left(2\pi e\sigma^2\right)\\
&\quad\left.-\left(\log A+\log\left(\frac{1-e^{-k}}{k}\right)+k\alpha-\frac{1}{2}\log\left(2\pi e\sigma^2\right)\right)\right\}\\
&=\lim_{A\to\infty}\left\{\mathcal{Q}\left(\frac{\log A}{\sigma}\right)\left(1+2\log\left(\sqrt{2\pi}\sigma\right)\right)+\frac{\log A}{\sqrt{2\pi}\sigma}e^{-\frac{(\log A)^2}{2\sigma^2}}\right.\\
&\quad\left.+\log\left(\frac{e^{k\frac{\log A}{A}}-e^{-k(1+\frac{\log A}{A})}}{1-e^{-k}}\right)\right\}\\
&=0+\log\left(\frac{1-e^{-k}}{1-e^{-k}}\right)+\log 1=0. \qquad \text{(B.47)}
\end{aligned}
$$

Now, we have to verify Condition (B.46). From (B.42) and (B.44) we derive:

$$
\lim_{A\to\infty}\frac{\mathrm{E}_{g,\max}[Y]}{A}=\alpha. \qquad \text{(B.48)}
$$

Therefore, with Condition (B.35) and the substitution (B.39) we obtain:

$$
\begin{aligned}
\alpha &= \lim_{A\to\infty}\frac{1}{k}-\frac{e^{-k\left(1+\frac{\log A}{A}\right)}\left(1+\frac{\log A}{A}\right)+e^{k\frac{\log A}{A}}\frac{\log A}{A}}{e^{k\frac{\log A}{A}}-e^{-k\left(1+\frac{\log A}{A}\right)}}\\
&= \frac{1}{k}-\frac{e^{-k}}{1-e^{-k}}. \qquad \text{(B.49)}
\end{aligned}
$$

□

Appendix C

Proofs of Chapter 5

C.1 Upper Bound for the Capacity of the Peak Power Constrained Channel using the Capacity Achieving Output Density

The capacity achieving output density $R^*(y)$ for an AWGN channel with variance σ^2 for an input in the interval $[0, A]$ has been found by Smith [10]:

$$R^*(y) = \frac{1}{2\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} + \frac{1}{2\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}. \tag{C.1}$$

Using this density in the first term of (2.17) we obtain:

$$\begin{aligned}
&\mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} W(y|X)\log R^*(y)\mathrm{d}y\right] \\
&= \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \log\left(\frac{1}{2\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}} + \frac{1}{2\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}\right)\mathrm{d}y\right] \\
&\leq \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \frac{1}{2}\log\left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}\right)\mathrm{d}y\right] \\
&\quad + \mathrm{E}_{Q^*}\left[-\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-X)^2}{2\sigma^2}} \frac{1}{2}\log\left(\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-A)^2}{2\sigma^2}}\right)\mathrm{d}y\right] \\
&= \log\left(\sqrt{2\pi}\sigma\right) + \mathrm{E}_{Q^*}\left[\frac{X^2+\sigma^2}{4\sigma^2}\right] + \mathrm{E}_{Q^*}\left[\frac{X^2+\sigma^2-2AX+A^2}{4\sigma^2}\right] \\
&\leq \log\left(\sqrt{2\pi}\sigma\right) \frac{\frac{A^2}{2}+\sigma^2+\frac{A^2}{2}+\sigma^2-A^2+A^2}{4\sigma^2} \\
&= \log\left(\sqrt{2\pi}\sigma\right) + \frac{1}{2} + \frac{A^2}{4\sigma^2},
\end{aligned} \tag{C.2}$$

where in the first inequality we used Jensen's inequality and in the second inequality $\mathrm{E}[X^2] \leq A \cdot \mathrm{E}[X]$ and Theorem 3.3. Then, we get the following

upper bound for the capacity:

$$C \leq \frac{A^2}{4\sigma^2}. \tag{C.3}$$

Appendix D

Mathematical Preliminaries

D.1 The $\mathcal{Q}(\cdot)$-function

A random variable Y is said to be Gaussian with mean m and variance σ^2, if the probability density is given by:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-m)^2}{2\sigma^2}}. \tag{D.1}$$

The cumulative distribution function of the random variable Y with density $f_Y(y)$ is defined as:

$$F_Y(y) = Pr[Y \leq y] = \int_{-\infty}^{y} f_Y(t)\,\mathrm{d}t. \tag{D.2}$$

For a Gaussian random variable it does not exist a closed-form expression of the cumulative distribution function. Instead the $\mathcal{Q}(\cdot)$-function is defined:

$$\mathcal{Q}(y) = \int_{y}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,\mathrm{d}t. \tag{D.3}$$

Then, the cumulative distribution function for the Gaussian random variable in (D.1) is given by:

$$F_Y(y) = 1 - \mathcal{Q}\left(\frac{y-m}{\sigma}\right). \tag{D.4}$$

Proof: Using the variable transformation $z = \frac{y-m}{\sigma}$ we obtain:

$$\begin{aligned} F_Y(y) &= \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(t-m)^2}{2\sigma^2}}\,\mathrm{d}t = 1 - \int_{y}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(t-m)^2}{2\sigma^2}}\,\mathrm{d}t \\ &= 1 - \int_{\frac{y-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,\mathrm{d}z = 1 - \mathcal{Q}\left(\frac{y-m}{\sigma}\right). \end{aligned} \tag{D.5}$$

□

Two important properties of the $\mathcal{Q}(\cdot)$-function listet in [12] are:

$$\mathcal{Q}(0) = \frac{1}{2}, \tag{D.6}$$

$$\mathcal{Q}(-y) = 1 - \mathcal{Q}(y). \tag{D.7}$$

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